How Long Does a Gold Rush Last?

Labour, Land and Machines after Africa's Gold Boom

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Abstract

Artisanal gold mining spread across Africa as the gold price rose until 2012. How long does such a gold rush last? I exploit the interaction between geology suitable for artisanal gold mining and the rise, fall and second rise of the world gold price, using household surveys from 33 African countries, satellite data and trade data. The jobs last as long as the price: men's farming in gold-suitable places falls by 16 percentage points at the 2012 peak and recovers 9.5 of them when the price drops. When the price climbed past its earlier peak after 2019, the work did not return, while forest loss grew further, mostly away from mapped mines. Imports of excavators into gold-exposed countries stayed high after the peak, while mercury, the input of manual mining, followed the price. School attendance in gold-suitable places stays lower after the peak, but the cohorts who were teenagers during the boom completed as much schooling as earlier ones, and households kept two thirds of their gain in wealth.

JEL codes: I25, J24, O13, O17, Q23, Q32.

Keywords: artisanal mining, gold, resource booms, mechanisation, schooling, Africa.

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1 Introduction

Between 2000 and 2012, the world price of gold rose from $279 to $1,670 an ounce, and across Africa people moved into artisanal mining, a labour-intensive and largely informal activity on which 130 to 270 million people depend worldwide (Girard et al., 2025). Then the price fell. By 2015 it was 30 percent below its peak, and it stayed near that level until 2018, before rising again to a new record by 2024. What happened to the people the gold rush had drawn in, and to the places it had changed?

Recent work shows what the boom did to gold-suitable places: it cleared forest and raised household wealth (Girard et al., 2025), lowered school attendance among adolescents (Ghazarian and Khan, 2025), raised women's employment in mining and trade and reduced violence by their partners (Byambasuren and Nguyen, 2026), and created new small towns (Girard and Pignède, 2026). A full price cycle asks how long these changes last. Artisanal mining needs little more than labour and tools, so people can leave it as cheaply as they entered, and work should follow the price in both directions. Other changes need not reverse. Cleared forest does not regrow when miners leave, a teenager who leaves school for the pits may not go back when they close, and assets bought in the boom outlast it. Who bears each change also differs, because men, women and children enter the gold economy in different ways.

Two features of the setting make the question hard to answer. Most African countries keep no record of where artisanal mining happens, and few household surveys code mining separately. And through most of the years that recent papers study, the gold price rose steadily: its correlation with the calendar year over 1990 to 2018 is 0.83, so a differential trend in gold-suitable places would look like a response to the price. The fall after 2012 and the rise after 2019 separate the two, because a trend continues in one direction while a response to the price turns with it.

In this paper, I follow work, land use, schooling and household wealth in gold-suitable places through the full price cycle. Following Girard et al. (2025), I measure exposure by the share of each half-degree cell's land on geology suitable for artisanal gold mining, and I compare cells within the same country and year as the lagged world price moves. Geology predates the boom and no African mining area sets the world price, so the variation is plausibly exogenous. I compare the peak of 2012 to 2014 with the years of low prices that followed, in the eight countries surveyed on both sides of the peak, and the years of low prices with the second rise of 2019 to 2024.

I build the household data from the Demographic and Health Surveys: 166 surveys from 33 countries between 1986 and 2025, in which 1.6 million women, 0.7 million men and 1.6 million households are linked to their half-degree cell through the survey's cluster coordinates. The data include the men's own interviews, which record each man's occupation, the surveys fielded after 2018, household rosters with school attendance and completed schooling, the survey's wealth index and the birth histories of mothers.

Forest loss, night lights and the footprint of mines come from satellite data, and imports of mining machines and mercury from international trade data.

The findings follow the cycle. First, the jobs last as long as the price, and farming carries the clearest signal. Across all countries, farming in gold-suitable places falls as the price rises, by about 4 percentage points over the rise to the peak in a fully gold-suitable cell, and men and women leave it in equal measure. In the eight countries surveyed on both sides of the peak, men's farming is 16.1 points lower at the peak than before the boom, on a mean of 41, and it recovers 9.5 points when the price falls. The men go into manual work, where most artisanal miners are recorded: it is 5.6 points higher at the peak, 29 percent of its mean, and gives up 4.9 of those points afterwards. Women move into trade, and their sales work rises by 5.5 points and gives up 4.0. Fewer women work at all, by 8 points at the peak on a mean of 57, and it recovers only 2 of those points, imprecisely.

Second, the second rush brings no jobs back. Surveys fielded in 2019 to 2024 interviewed people when the lagged price was 13 percent above its level in the peak surveys. Had men in these ten countries responded as they did across the peak, their manual work would have risen by about 9 points; the estimate is -0.2, and the 95 percent interval excludes any rise above 3.2. Men's farming does not fall and women's sales do not return. The result holds without the surveys of 2020 and 2021, without any single country, and among men aged 15 to 24, who were children at the peak.

Third, the land does not recover. Forest loss rises with the price, and at the peak it is about 22 percent higher in a fully gold-suitable cell than in 2001 to 2005. It does not fall when the price drops, and in the second rise it grows by a further 6 percent. That further clearing lies away from the mines. It occurs in cells with no record of gold or artisanal mining and with no mine on a satellite map of mine footprints (Tang and Werner, 2023), and on a global map of the drivers of forest loss most of the extra clearing lies in pixels attributed to farming, shifting cultivation and logging. Gold-suitable land kept being cleared while the workers stayed away.

Fourth, machines arrived, but they do not account for the cleared land. Relative to a country with no gold-suitable land, a country at Ghana's share of 0.42 imported about 2.4 times its earlier value of excavators by the peak, and these imports stayed high after it. Mercury, the input of manual amalgamation, instead rose and fell with the price, as men's manual work did. Imports of road rollers, a construction machine, also rose after the peak, so the excavators that stayed may have built roads as well as dug pits.

Fifth, the boom leaves a longer trace on places than on work. Compared with children aged 6 to 11 in the same cell and survey, whom school closures or mercury would affect as well, attendance at ages 16 to 20 is 5.5 points lower at the peak, on an attendance rate of 47 percent. Half of the gap remains after the peak and through the second rise, although the young men who left farming return to it. The cohorts who were teenagers during the boom nonetheless catch up: among adults aged 21 to 45,

years of schooling do not fall with the price their cohort faced at ages 12 to 20, and the 95 percent interval excludes a loss larger than 0.36 years on a mean of 5.9. Households keep about two thirds of a rise in the wealth index of 0.29 standard deviations, and night lights keep rising after the peak. Child mortality does not change on average; it rises only in towns that predate the boom.

Before turning to the data, I note three limits of this evidence. First, the cycle in work rests on the eight countries surveyed on both sides of the peak. When inference treats each country as one cluster, only the fall in men's farming at the peak is statistically significant at the 5 percent level, so I put more weight on what recurs.[1] Dropping each country in turn leaves the fall in men's manual work after the peak between 3.2 and 5.7 points, the five countries that enter both comparisons show the same cycle, and in 1,000 reassignments of gold suitability across cells within each country, no more than 2 percent of draws produce a cycle as large as the actual one. Second, part of the reversal is common to whole gold regions: with region-by-year effects, men's farming still falls and recovers, but the fall in manual work and trade after the peak is small. Third, none of the data observe machines at mine sites: the trade data are national, and the satellite map of mine footprints is a single snapshot taken after the peak. Still, a full price cycle gives a test that a rising price cannot, of whether each change follows the price back down, and work, land and schooling give different answers.

These results add to work on local resource booms. Large mines raise local incomes through their demand for inputs (Aragón and Rud, 2013) and move women from farming into services (Kotsadam and Tolonen, 2016), and Huang et al. (2026) find that higher mineral prices shift work from agriculture to services around mines worldwide. Evidence on what happens when a boom ends comes mainly from the United States, where wages and employment in oil and gas counties rise in booms and fall in busts (Allcott and Keniston, 2018), and incomes in the boomtowns of the American West fell below their earlier path once the boom ended (Jacobsen and Parker, 2016). I show that artisanal gold work in Africa reverses with the price while the land use and much of the wealth it brought do not, and that a later boom raised forest clearing without bringing the workers back.

The schooling results join work on how local labour demand shapes education. Export factories in Mexico raised school dropout (Atkin, 2016), the fracking boom in the United States reduced schooling among young men (Cascio and Narayan, 2022), and the US housing boom lowered college attendance in ways that persisted after the bust (Charles et al., 2018). Ghazarian and Khan (2025) show that the rise in the gold price lowered school attendance among adolescents near gold-suitable land. I find that lower attendance in gold-suitable places outlasts the work that caused it, since only half of the lost attendance returns when the price falls, but that the cohorts who were

[1] Standard errors are clustered by cell. With eight country clusters, the wild cluster bootstrap gives a p-value of 0.02 for the fall in men's farming at the peak and of 0.16 to 0.30 for the reversals after it.

teenagers during the boom complete as much schooling as earlier ones.

The closest papers use the same geology and the world gold price. Girard et al. (2025) show that the boom raised deforestation and household wealth, Byambasuren and Nguyen (2026) that it raised women's employment and reduced violence by their partners, and Girard and Pignède (2026) that it created small towns without structural change. In Ghana, Duodu and Danquah (2026) combine gold-bearing Birimian geology with the gold price and find that fully gold-suitable districts host 19 percent fewer formal firms. Bazillier and Girard (2020) compare the two kinds of mining in Burkina Faso and find that household consumption rises with the gold price near artisanal mines but not near industrial ones. Each studies a rising price. I follow the same places through the fall that followed the peak and the rise that followed the fall, and I find that work, land, schooling, wealth and health respond to the cycle on different clocks.

Section 2 describes artisanal mining and the price cycle. Sections 3 and 4 describe the data and the empirical strategy. Section 5 reports the results and Section 6 their robustness. Section 7 concludes.

2 Artisanal Gold and the Price Cycle

2.1 Artisanal Gold Mining in Africa

"Artisanal and small-scale mining refers to formal or informal mining operations with predominantly simplified forms of exploration, extraction, processing, and transportation" (OECD, cited in Girard et al., 2025). It employs an estimated 44.75 million people worldwide (World Bank, 2020), and in Africa its main mineral is gold (Girard et al., 2025). The tools are simple. Miners dig pits and shafts, or wash river gravel, with picks, shovels, pans and sluices, and they often separate the gold with mercury. Most work close to home, much of it in the dry season between harvests (Bazillier and Girard, 2020), and much of it without a licence.

Men do most of the digging. Women are a large share of the workforce, estimated at 40 to 50 percent of artisanal miners in Africa, and many work in processing, including amalgamation with mercury (Hinton et al., 2003). Adolescents work on sites as well, and school attendance near gold-suitable land falls when the gold price rises (Ghazarian and Khan, 2025). Few household surveys record this work under its own name. In the occupation codes of the Demographic and Health Surveys, most diggers appear as manual workers.

Industrial gold mines differ in scale and in their links to the local economy. In Burkina Faso in 2014, an estimated 640,800 people worked as gold diggers, while the industrial mining sector reported 6,464 employees (Bazillier and Girard, 2020). Industrial mines are capital-intensive projects licensed by the state, run by international companies with skilled employees on formal contracts, and they import most of their equipment

(Bazillier and Girard, 2020); their local effects run mainly through their purchases of local inputs (Aragón and Rud, 2013). The two kinds of mining often sit on the same geology: of the 139 half-degree cells that ever host an industrial gold mine, 133 contain land suitable for artisanal mining.

Artisanal mining has also changed its technology. In Ghana, tens of thousands of Chinese miners arrived from the late 2000s (Hilson et al., 2014) and brought excavators and large wash plants to artisanal sites, which turned over far larger areas of land than digging by hand (Crawford and Botchwey, 2017). Where machines do the digging, a rise in the price can raise output and the land cleared without a matching rise in employment. The machines also drew the state in. In 2017 Ghana suspended small-scale mining, and soldiers sent to enforce the ban burned excavators at illegal sites (Hilson, 2017).

2.2 The Gold Price Cycle

After two decades of decline, the world price of gold stood at $279 an ounce in 2000. It rose from 2001, nearly tripled between 2005 and 2010 and peaked at an annual average of $1,670 in 2012. Then it fell. By 2015 it was 30 percent lower, at $1,161, a fall of 22 percent in real terms, and it stayed close to that level until 2018. The second rise began in 2019: the price reached $1,770 in 2020 and $2,388 in 2024, 43 percent above its 2012 level, or 37 percent in real terms (World Bank Pink Sheet).

Figure 1 shows the lagged price index that I use, with 2013 equal to one, and the five windows that organise the analysis. The index averages 0.23 in 2000 to 2005, the reference period; 1.10 at the peak, 2012 to 2014; 0.87 after the peak, 2015 to 2018; and 1.18 in the second rise, 2019 to 2024. The rise between 2006 and 2011 and the years before 2000 complete the path. African mining areas take the world price as given, and the geology that decides where artisanal gold can be mined predates the boom.

Household surveys cover the cycle unevenly (Figure 1, Panel B). Each country fields a survey only every four to six years, so only eight countries were surveyed both at the peak and after it: Benin, Chad, Guinea, Mali, Nigeria, Rwanda, Senegal and Zambia. Ten were surveyed both after the peak and in 2019 to 2024, five of them among the eight. The first group carries the comparison of the peak with the years after it; the second carries the comparison of those years with the second rise.

The setting suggests how each outcome should move over the cycle. Because artisanal mining needs little more than labour and tools, people can leave it as cheaply as they entered, and men's work should follow the price in both directions. Land cleared for pits does not regrow when miners leave, and assets bought during the boom outlast it. Where excavators replace diggers, a later rise in the price can raise output and the land cleared without drawing workers back. For teenagers, the boom raises the return to work against school; whether they return to school when the price falls depends on how hard it is to resume after leaving. Section 5 takes these outcomes in this order.

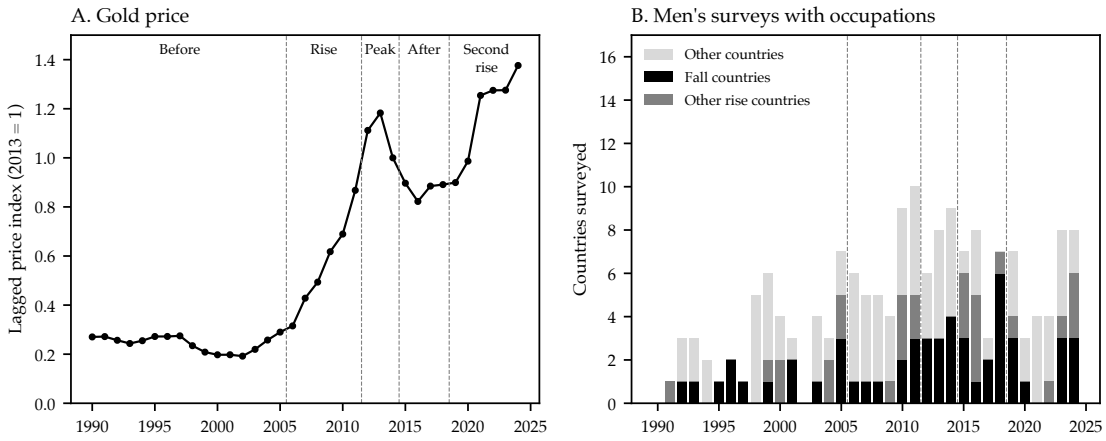


Figure 1—The gold price and the timing of the surveys

Notes: Panel A plots the gold price index lagged by one year (nominal, 2013 equal to one), with the windows used in Section 4. Panel B counts countries with a men's survey recording occupations in each interview year. Fall countries are the eight surveyed both at the peak and after it; other rise countries are those surveyed after the peak and in 2019 to 2024 that are not fall countries. Sources: World Bank Pink Sheet; Demographic and Health Surveys.

3 Data

I combine a map of geology suitable for artisanal gold mining, household surveys that I assemble and link to the map, and satellite data on forest loss. Appendix A describes the construction of each file.

3.1 Gold-Suitable Land

Most African countries keep no register of artisanal mines, and where a site opens may depend on local labour supply. Following Girard et al. (2025), I measure where artisanal gold could be mined. Their map classifies bedrock across Africa as suitable for artisanal gold mining by its age, chemistry and structure, and it marks about 18 percent of the continent (Figure 2, Panel A). For each of the 10,686 half-degree PRIO-GRID cells of the continent, I compute the share of its area on that bedrock. The share is positive in 4,514 cells, where it averages 0.43 (Panel B). Exposure is this share times the world gold price index lagged by one year: it is zero in a cell without suitable land in every year and rises with the price in a cell with it. The map captures where artisanal mining takes place. Other records of gold or artisanal gold mining, from exploration surveys, administrative and survey records of sites, mercury emissions and sites detected in satellite images, exist in 44 percent of cells with suitable land and in 8 percent of the others (Panel C). On the satellite map of mine footprints of Tang and Werner (2023), a fully gold-suitable cell with no industrial gold mine is 5.4 percentage points more likely to contain a mine than a cell without suitable land, against an average of 2.7 percent, and the artisanal mercury emissions that Steenhuisen and Wilson (2019) allocate across space are higher there as well. Of the 139 cells that ever host an industrial gold mine,

133 also contain suitable land. Section 6 therefore adds an industrial-mine indicator times the same price index to every main estimate.

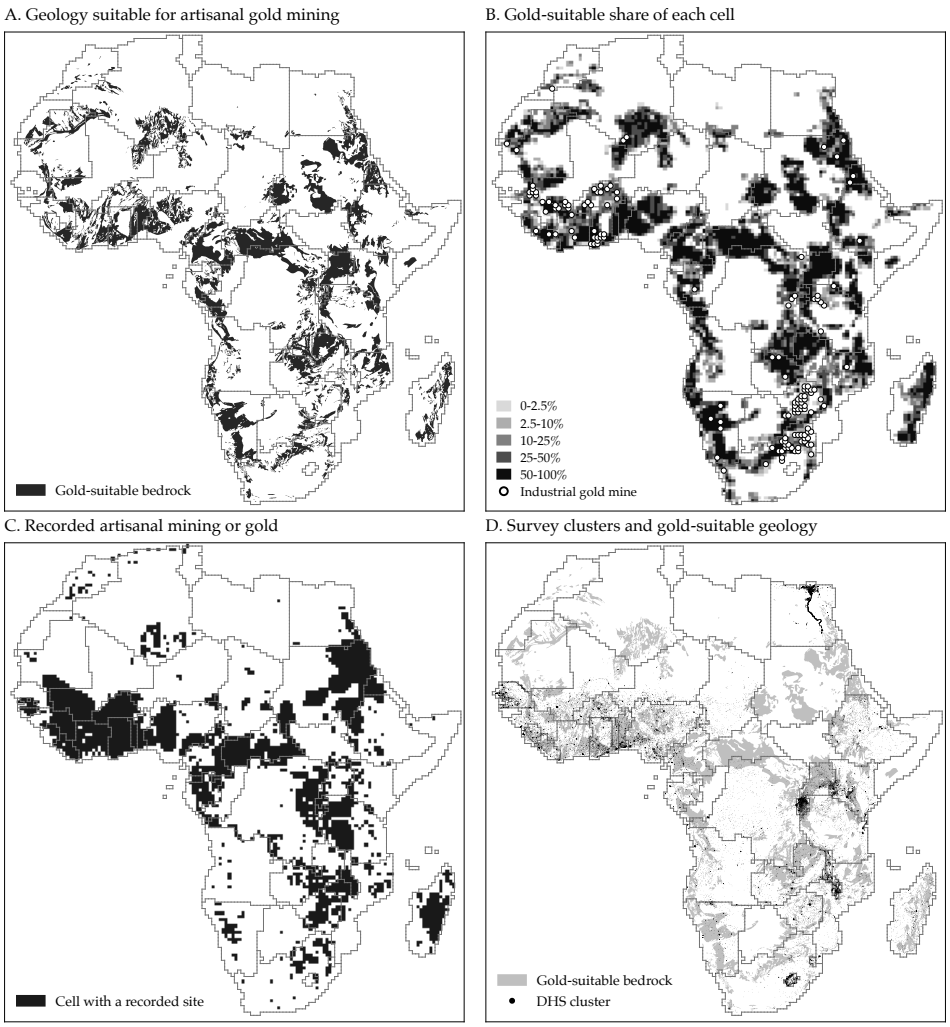


Figure 2—Where artisanal gold can be mined

Notes: Panel A maps the bedrock that Girard et al. (2025) classify as suitable for artisanal gold mining. Panel B shades each 0.5-degree cell by the share of its land on that bedrock, which is the exposure measure; white cells have none, and circles mark cells that ever host an industrial gold mine. Panel C marks cells with any other record of gold or artisanal gold mining in the data of Girard et al. (2025): exploration records, gold deposits, administrative and survey records of artisanal sites, mercury emissions and artisanal sites detected in satellite images. Such records exist in 44 percent of cells with gold-suitable land and in 8 percent of other cells. Panel D overlays the 65,746 survey clusters of the Demographic and Health Surveys used in the paper on the bedrock of Panel A; cluster coordinates are displaced by up to 5 kilometres in rural areas and, for a few, up to 10.

3.2 Household Surveys

The household data come from the Demographic and Health Surveys (ICF, 2025), which I assemble from the raw files of every standard survey in 33 African countries: 166 surveys fielded between 1986 and 2025. Each survey is a repeated cross-section, so I follow places, not people. The 132 surveys that record the coordinates of their sampling

clusters locate 65,746 clusters, and I assign each cluster to its half-degree cell (Figure 2, Panel D); 99 percent of interviewed people in these surveys are linked to a cell. To protect respondents, DHS displaces urban clusters by up to 2 kilometres and rural clusters by up to 5 kilometres, and a few by up to 10. Displacement adds noise to exposure without shifting it toward gold-suitable land, and 26 percent of clusters lie close enough to a cell edge to have crossed it. I flag them and drop them in Section 6.

Working from the raw files gives three things that the survey data of earlier studies lack: the men's interviews, in which each man reports his own occupation; the surveys fielded after 2018; and the household rosters, which record school attendance for every member.[2]

3.3 Outcomes

Work comes from the occupation each respondent reports, coded into the survey's standard groups. Each work and attendance outcome is an indicator equal to one if the person works in the sector or attends school, which I multiply by 100, so that coefficients are in percentage points. Manual work is skilled or unskilled manual work, where most artisanal miners are recorded, and agriculture includes farming on one's own account and as an employee. In the surveys whose detailed occupation codes name mining without pooling it with other trades, I also measure mining directly. For men, the outcomes are their own occupations: 664,621 men in 119 surveys from 32 countries. For women, I use their own occupation and whether they currently work, for 1.64 million women in 130 surveys, and the measure of Girard et al. (2025), which equals one if the woman or her partner works in the sector. School attendance is attendance at any time in the current school year, recorded for every member of the household. Household wealth is the survey's index of assets and housing, which DHS standardises within each survey; I compare it within surveys only, for 1.36 million households in 101 surveys from 2003. Table 1 reports the means of each outcome.

3.4 Forest Loss

Forest loss comes from Global Forest Change, version 1.12, which maps tree cover in 2000 and its loss in each year from 2001 to 2024 at a resolution of 30 metres (Hansen et al., 2013). I aggregate it to cells as the area of canopy above 30 percent in 2000 and the area of that forest lost each year, and I keep the 4,627 cells with at least one square kilometre of forest in 2000. Measured loss across Africa doubles between 2012 and 2014, in part because the product detects loss better from those years. Country-by-year effects absorb the common change, and Section 6 lets it differ with each cell's forest cover. Night lights come from the harmonised series of Li et al. (2020), which calibrates

[2]Ethiopian surveys before 2024 record dates in the Ethiopian calendar, which I convert to the Gregorian calendar. In six surveys more than 40 percent of couples have no sector recorded, and I treat their occupations as missing.

Table 1—Summary statistics

	All	Gold-suitable cells	Other cells
Panel A: Men, own occupation			
Gold-suitable share of cell	0.28	0.45	0.00
Manual work (%)	20.0	19.4	20.9
Agriculture (%)	39.1	41.4	35.5
Mining, named in occupation (%)	0.81	0.93	0.59
Age	30.9	30.8	31.1
Men	664,621	409,939	254,682
Surveys	119	115	112
Panel B: Women			
Currently working (%)	54.4	61.2	46.0
Agriculture, own occupation (%)	25.4	32.5	16.5
Manual work, own occupation (%)	7.2	8.0	6.4
Manual work, woman or partner (%)	20.7	19.5	22.3
Women	1,640,605	909,101	731,504
Surveys	130	126	123
Panel C: Children and households			
Attending school, ages 12–15 (%)	75.8	77.8	73.2
Attending school, ages 16–20 (%)	46.5	46.5	46.5
Children aged 12–20	1,454,629	804,361	650,268
Wealth index (within-survey z-score)	0.00	−0.04	0.06
Bottom two national wealth quintiles (%)	41.1	42.4	39.4
Households	1,361,516	796,054	565,462
Panel D: Cells with forest in 2000, 2001–2024			
Tree cover in 2000 (km^2)	1272	1413	1105
Forest loss (% of 2000 cover a year)	0.57	0.59	0.54
Cells	4,627	2,505	2,122

Notes: Means. Gold-suitable cells have any land on the bedrock of Figure 2. Panels A to C are drawn from the 132 Demographic and Health Surveys with cluster coordinates, 1986 to 2025; the count of surveys differs by panel because not every survey has a men's file or records occupations. Occupations follow the survey's standard groups: manual is skilled and unskilled manual work, agriculture includes agricultural employees. Mining is the share of men whose survey-specific occupation names mining, quarrying or gold digging without pooling it with other trades, in the surveys that code it. The woman-or-partner measure follows Girard et al. (2025). Attendance is attendance at any time in the current school year. The wealth index is the survey's asset index, standardised within each survey. Forest loss is from Hansen Global Forest Change, version 1.12.

DMSP images for 1992 to 2013 and converts VIIRS images for 2014 to 2024 to the same scale; I average them over each cell.

Table 1 compares cells with and without gold-suitable land. Men in gold-suitable cells farm more (41 against 36 percent), and women there are more likely to work, mostly in agriculture (32.5 against 16.5 percent). Men name mining as their occupation in 0.9 percent of cases in gold-suitable cells and 0.6 percent elsewhere, a small share because most miners report another occupation. Attendance at ages 16 to 20 is the same in both groups, and wealth is slightly lower in gold-suitable cells. These are differences in levels, which cell fixed effects absorb; the design compares the same cells as the price moves.

4 Empirical Strategy

I compare cells within the same country and year that have more or less gold-suitable land, as the world price of gold moves. For person or household i in cell c, country k and year t, I estimate

$$Y_{ickt} = \beta \left(S_c \times P_{t-1} \right) + X'_{ickt}\gamma + \alpha_c + \lambda_{kt} + \varepsilon_{ickt}, \tag{1}$$

where S_c is the share of the cell on gold-suitable bedrock, P_{t-1} is the world gold price index in the year before the interview, with 2013 equal to one. The coefficient β measures how much more the outcome moves with the price in a fully gold-suitable cell than in a cell with no suitable land. Between 2000 to 2005 and the peak of 2012 to 2014, the index rose by 0.87, so 0.87β is the effect of the boom on a fully gold-suitable cell.

To follow the price through its cycle, I replace $S_c \times P_{t-1}$ with the suitable share interacted with indicators for the windows of Figure 1:

$$Y_{ickt} = \sum_{w \in \mathcal{W}} \beta_w \left(S_c \times \mathbf{1}[t \in w] \right) + X'_{ickt}\gamma + \alpha_c + \lambda_{kt} + \varepsilon_{ickt}, \tag{2}$$

where $\mathcal{W}$ contains the 1990s, the rise (2006 to 2011), the peak (2012 to 2014), the years after the peak (2015 to 2018) and the second rise (2019 to 2024), and 2000 to 2005 is the reference period. The reference is the trough of the price, when the lagged index averaged 0.23 and artisanal gold paid least, so each coefficient measures how far a phase of the cycle moved gold-suitable places from their position before the boom; the 1990s, with fewer surveys and a slightly higher price, enter as a separate window. Each β_w is the change in the outcome in a fully gold-suitable cell between the reference period and window w. Three comparisons carry the results. The peak coefficient measures the boom. The difference between the coefficients after the peak and at the peak measures the response to the fall, estimated in the eight countries surveyed on both sides of the peak. The difference between the second rise and the years after the peak measures the

response to the second rise, estimated in the ten countries surveyed in both windows. The 1990s coefficient shows whether gold-suitable cells were already moving apart from other cells before the price rose.

Cell effects α_c absorb every fixed difference between places, including the differences in levels in Table 1 and the direct effect of geology. Country-by-year effects λ_{kt} absorb national shocks and policies and the effect of the gold price on each national economy, so the main effect of the price is not identified and does not appear. The price enters with a lag because survey questions on work refer to the past year and because a change in the world price takes time to reach the price paid at a site. The controls X_{ickt} are age and its square for individuals and, following Girard et al. (2025), extreme temperature and the cell's suitability for its main cash crop times that crop's lagged world price. The two cell controls end in 2018, so estimates that use the surveys after 2018 omit them, and Section 6 reports the earlier estimates with and without them. Industrial gold mines sit on the same geology, in 133 of the 139 cells that host one, so β is the response of gold-suitable places as a whole, including any response of industrial mines there. I leave industrial mines out of equation (1) because some opened during the boom, so their presence partly responds to the price; Section 6 adds mines producing by 2000, and then every industrial mine, each interacted with the price.

The same equation applies to each outcome, with the unit adapted. For men's and women's own work and for school attendance, the unit is the person. For the measure that counts the woman's or her partner's work, I weight each woman by the inverse of the number of interviewed women in her household, so that each household counts once; co-wives would otherwise contribute the same husband's occupation several times. For wealth the unit is the household, and for forest loss it is the cell and year. I cluster standard errors by cell, the level at which exposure varies.[3]

Identification rests on one assumption: had the gold price not changed, outcomes in more and less gold-suitable cells of the same country would have followed the same trends. No African mining area sets the world price of gold, and the geology that decides where artisanal gold can be mined predates the boom, so neither side of the interaction responds to local conditions. The assumption could still fail if gold-suitable places were on different paths for other reasons. Because the price rose steadily for a decade, a differential trend would look like a response to the boom, which is why the results follow the price down after 2012 and up again after 2019: a trend continues in one direction, while a response to the price turns with it.

Surveys sample different people in each round, and a boom may change who lives in gold-suitable cells. Men report their own occupations, so changes in household structure do not move the men's results, and the results for teenagers compare them with younger children in the same cell and survey; Section 6 also re-estimates the cycle

[3]The comparisons of the peak with the years after it rest on eight countries. For these, I also report p-values from a wild cluster bootstrap with country clusters, which enumerates all 2^8 sign patterns (Section 6).

for respondents who have lived in the place for at least five years. Displacement of the survey coordinates places some clusters in the wrong cell, which biases the estimates toward zero, so Section 6 drops the clusters near a cell edge as well. Because the comparisons across the peak rest on eight countries, I report each result without each country in turn.

5 Results

5.1 Farming Follows the Price

In Table 2, I report the average effects of the gold price on work, from equation (1) in all countries with surveys up to 2018, for all adults and for men and women separately. As the price rises, people in gold-suitable places leave farming. Agriculture falls by 4.9 percentage points per unit of the price index (Panel A, column 4), or by about 4.3 points over the rise to the peak, 14 percent of its mean of 31, and men and women leave it in equal measure. They go to different work. Women move into sales, by 3.8 points per unit of the index on a mean of 17 (Panel C, column 5), while men's sales do not change and services fall slightly for both (column 6). Manual work rises for both, by about 1 point per unit of the index, but imprecisely.

These averages pool the boom with the decades before it, and they cannot show what happened when the price fell. In Figure 3, I therefore follow all adults, men and women through the windows of the price cycle in the eight countries surveyed both at the peak and after it, with the price in each window drawn behind the estimates; Appendix Table B1 reports the coefficients.

The shifts in occupation reverse when the price falls, and farming moves most. At the peak, adults in a fully gold-suitable cell farm 12.9 percentage points less than in 2000 to 2005, and more of them work in manual jobs (2.2 points) and in sales (4.5 points). After the peak, farming recovers 5.2 points, and manual work and sales give up 2.4 and 2.8 points, so the jobs the boom created last only as long as the price.

The average hides two paths. Men carry the swing between farming and manual work (Panels A and B). At the peak, their farming is 16.1 points lower on a mean of 40.8, and their manual work is 5.6 points higher, 29 percent of its mean of 19.0. Of all the cycle estimates, this fall in farming is the one that remains statistically significant at the 5 percent level when inference treats each of the eight countries as a single cluster.[4] After the peak, farming recovers 9.5 of its 16.1 points and manual work gives up 4.9 of its 5.6, so manual work is back near its level before the boom and farming has recovered more than half of its fall. Only part of the manual response is recorded as mining. In the seven countries whose occupation codes name mining or quarrying, men's manual

[4]Wild cluster restricted bootstrap with country clusters, over all 256 Rademacher draws: $p = 0.02$ for men's farming at the peak and 0.13 for their manual work, and 0.16 and 0.18 for the recovery of farming and the fall of manual work after the peak.

Table 2—Artisanal gold and work: average effects

	Working (1)	Manual (2)	Extraction (3)	Agriculture (4)	Sales (5)	Services (6)
Panel A: All adults						
Suitable × lagged price	−1.32	1.05	0.29	−4.90***	2.98***	−0.95*
	(1.42)	(0.77)	(0.37)	(1.71)	(1.00)	(0.50)
Mean of outcome (%)	61.0	10.3	0.9	31.1	14.0	4.2
Observations	1,691,736	1,657,666	1,116,355	1,657,666	1,657,666	1,657,666
Panel B: Men						
Suitable × lagged price	−0.65	1.16	0.22	−5.04***	0.55	−0.61
	(1.73)	(1.11)	(0.63)	(1.91)	(0.73)	(0.59)
Mean of outcome (%)	77.3	18.4	1.9	41.2	7.6	5.2
Observations	474,271	475,389	344,150	475,389	475,389	475,389
Panel C: Women						
Suitable × lagged price	−1.43	1.40	0.53	−4.85**	3.84***	−1.06*
	(1.70)	(0.88)	(0.41)	(1.94)	(1.35)	(0.61)
Mean of outcome (%)	54.7	7.0	0.5	27.0	16.5	3.8
Observations	1,217,450	1,182,257	772,198	1,182,257	1,182,257	1,182,257

Notes: Each cell reports equation (1) for an indicator equal to 100 if the respondent's own occupation is in the sector and 0 otherwise, so that coefficients are in percentage points, in all countries with surveys up to 2018. The coefficient is the change in a fully gold-suitable cell as the lagged price index rises by one, in percentage points; the index rose by 0.87 between 2000–05 and the peak, so 0.87 times a coefficient is the effect of the boom (divide by 100 for the units of Girard et al. (2025)). Panel A stacks the men's and women's interviews and controls for sex. Extraction is manual work whose occupation label names mining or quarrying, in the surveys whose labels code it. All regressions control for temperature, the crop-price index, age and age squared, with cell and country-by-year effects. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

work rises by 6.9 points at the peak and falls by 5.3 after it, while extraction rises by 1.2 points on a mean of 1.3 (Appendix Table B1, column 3). Extraction nearly doubles, yet it is about a fifth of the manual response; the rest appears under other manual labels, consistent with diggers whom the surveys record as manual workers (Section 2).

Women follow a different path. They leave farming at the peak, by 11.0 points on a mean of 25, but the jobs they take are in sales (Panel C), up 5.5 points at the peak and down 4.0 after it, and their manual work does not change. Services fall slightly for everyone during the boom (Panel D). Fewer women work at all: the share of women in any work is no different in the 1990s, falls by 7.5 points during the rise and by 8.1 points at the peak, 14 percent of its mean of 57, and recovers only 2.1 of those points after the peak, an estimate whose 95 percent interval reaches 5.9. I return to women's work in the second rise in Section 5.2.[5]

Before the boom, the 1990s coefficients are imprecise. Farming in gold-suitable cells

[5] Byambasuren and Nguyen (2026) find that women's employment in mining and in retail rises with the gold price. They measure employment by industry, for ever-married women and the husbands they report, while I measure each respondent's own occupation and current work; both patterns can hold if the women who keep working move toward mining and trade while others stop working.

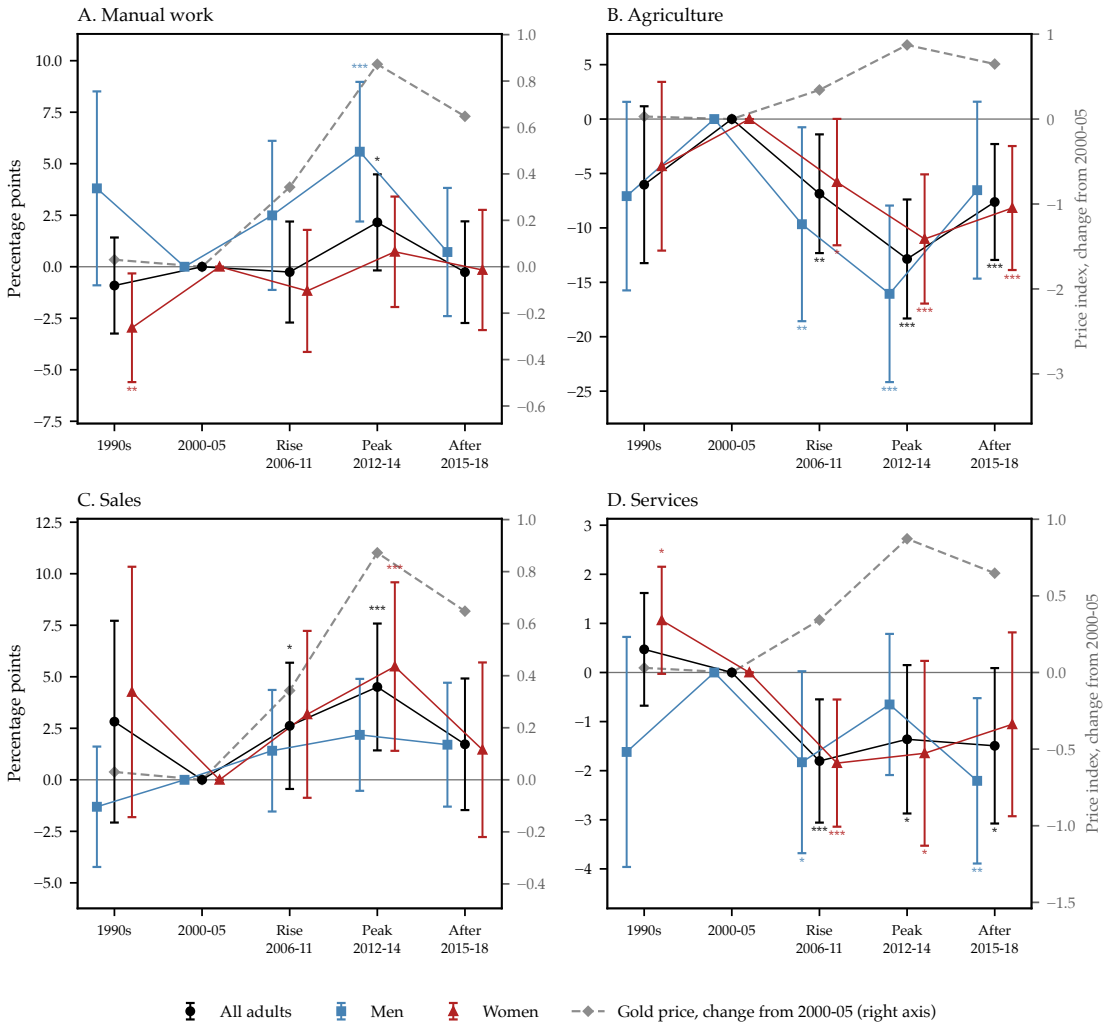


Figure 3—Work over the gold price cycle

Notes: Coefficients of equation (2) for manual work (Panel A), agriculture (Panel B), sales (Panel C) and services (Panel D), for all adults, men and women, each respondent's own occupation as an indicator equal to 100 or 0, so that coefficients are in percentage points, in the eight countries surveyed both at the peak and after it, surveys to 2018, relative to 2000 to 2005; Appendix Table B1 reports the estimates. The dashed line (right axis) is the mean lagged gold price index in each window minus its mean in 2000 to 2005, the trough of the price (2013 equal to one): 0.03 in the 1990s, 0.34 during the rise, 0.87 at the peak and 0.65 after it. Figure 4 reports the second rise. Bars are 95 percent confidence intervals from standard errors clustered by cell.

was 4 to 7 points lower than in 2000 to 2005, and women's manual work 3 points lower. In both halves of the decade the lagged price index was within 0.04 of its mean in 2000 to 2005, so I also split the 1990s into the surveys of 1990 to 1995 and those of 1996 to 1999.[6] None of the twelve estimates for all adults' farming and manual work, men's farming and manual work and women's sales and current work is statistically significant at the 5 percent level. In neither half of the decade was farming more than 3 points higher

[6]The early surveys are Nigeria 1990, Senegal 1992 and 1993 and Mali 1995; the late ones are Benin and Mali 1996, Senegal 1997 and Guinea 1999. Men's farming is 2.9 points higher in 1990 to 1995 and 8.7 points lower in 1996 to 1999 than in 2000 to 2005, and their manual work is 5.6 and 3.5 points higher.

than in 2000 to 2005, so gold-suitable places were not already leaving farming before the price rose. Section 6 pools the 1990s with 2000 to 2005 as the reference. The responses to the fall compare the peak with the years after it, so they do not depend on the reference period.

The censuses do not show the move out of farming. In Online Appendix OA3, I estimate equation (1) on the IPUMS-International censuses of thirteen countries. Men's employment in mining rises with the price, by 0.8 points per unit of the price index on a mean of 1.8, but the share of workers in agriculture rises too, by 2.8 points on a mean of 45, where the surveys show a fall. The two sources measure different things. No country has a census both at the peak and after it, so the censuses cannot follow the cycle, and their units are districts or regions, several times larger than a cell, in which gold-suitable land is a smaller share of where people live. I therefore treat the censuses as evidence that the mining response reaches official employment counts, and not as a test of the farming result.

5.2 The Second Rise Brings No Jobs Back

From 2019, the price rose again, and the surveys of 2019 to 2024 interviewed men at a higher price than the peak surveys did: the lagged index averaged 1.28 at their interview dates, against 1.13 at the peak and 0.88 in 2015 to 2018. Had men responded to this rise as they responded to the peak, their manual work in a fully gold-suitable cell would have risen by about 9 points between the two windows.

In Figure 4, I plot the change from 2015 to 2018 to 2019 to 2024 in the ten countries surveyed in both windows, against the change that the response at the peak implies (Appendix Table B2 reports the estimates). Men's manual work does not move: the estimate is -0.2 points, and the 95 percent interval, from -3.5 to 3.2, rules out the rise of about 9 points that the response at the peak implies. Men's farming does not fall either, against an implied drop of 15 points. The absence holds without the interviews of 2020 and 2021, when COVID-19 disrupted fieldwork, without any single country, and among men aged 15 to 24, who were children at the peak; in every case the upper end of the 95 percent interval stays below 4.5 points. Women's trade does not return either. Had their sales responded as they did at the peak, they would have risen by 5.3 points; the estimate is -0.4, with an interval that ends at 2.3; among all adults, manual work and sales stay flat against implied rises of 3.6 and 3.1 points.

The fall and the second rise are measured in different countries, eight for the first and ten for the second, so the contrast could reflect which countries were surveyed. Five countries enter both groups: Mali, Nigeria, Rwanda, Senegal and Zambia. In these five alone, men's farming falls by 19.5 points at the peak and recovers 9.2 after it, men's manual work gives up 5.8 points after the peak, and women's sales give up 4.0. In the second rise, men's manual work changes by -1.0 point against an implied rise of 9.8, their farming by 3.6 against an implied fall of 14.7, and women's sales by 0.3 against

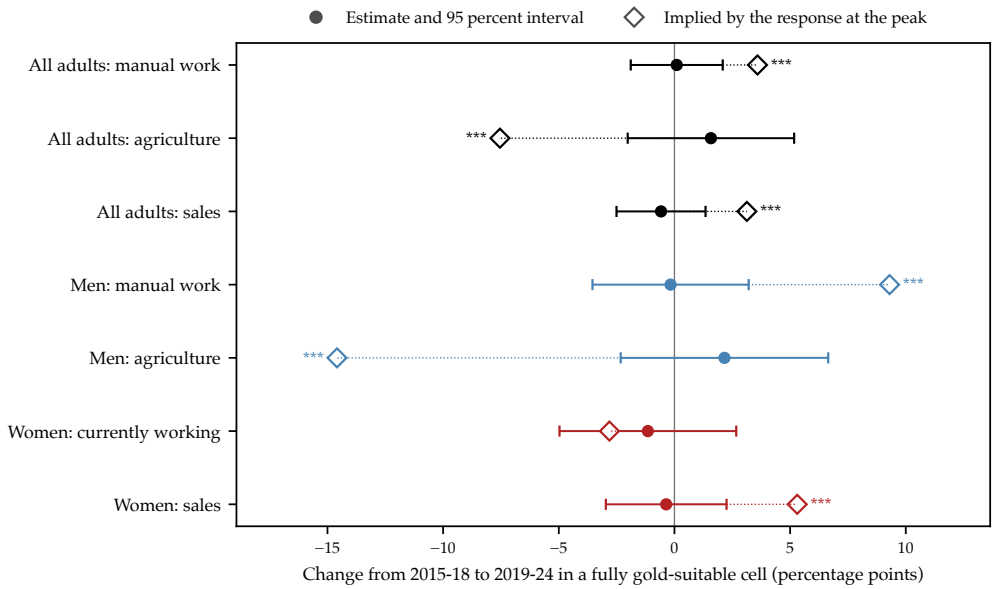


Figure 4—Work in the second rise of the gold price

Notes: Each row plots the difference between the 2019–24 and 2015–18 coefficients of equation (2) in a fully gold-suitable cell (filled circle, with its 95 percent confidence interval) and the change implied if the outcome had responded to the second rise as it responded to the peak (hollow diamond): the change from 2015–18 to the peak, scaled by the change in the mean lagged price index at the interview dates. Each respondent's own occupation, as an indicator equal to 100 or 0, in the ten countries surveyed both in 2015–18 and in 2019–24, surveys to 2024. Stars test the estimate against the implied change: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Appendix Table B2 reports the estimates. Standard errors clustered by cell.

an implied rise of 6.3. The same countries that left mining when the price fell did not return to it when the price rose again.

5.3 The Land Does Not Recover

The land follows a different path from the jobs. In Figure 5, Panel A, I plot equation (2) for the area of forest lost each year in all 4,627 forested cells of the continent. During the rise, a fully gold-suitable cell loses about 7 percent more forest a year than in 2001 to 2005, and at the peak about 22 percent more. After the price falls, the excess barely changes. Clearing continues at its peak rate in the years when men return to farming, so the land cleared during the boom keeps growing. Panel B allows the improvement in satellite detection after 2012 to differ with each cell's forest cover. The coefficients are smaller, but the shape is the same: loss rises with the boom and does not fall after the peak.

In the second rise, clearing speeds up again. Forest loss in a fully gold-suitable cell grows by a further 6 percent between 2015 to 2018 and 2019 to 2024 (Panel A), and by 4 percent when detection is allowed to change with forest cover (Panel B); both differences are statistically significant at the 1 percent level. The change from the peak to 2015 to 2018 is −1 percent, and its 95 percent interval excludes a fall larger than

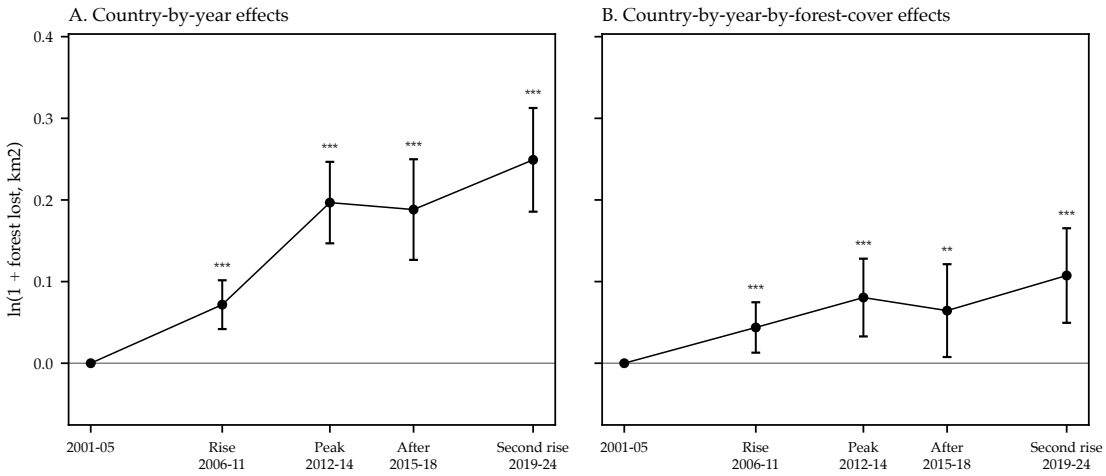


Figure 5—Forest loss over the gold price cycle

Notes: Coefficients of equation (2) for the logarithm of one plus the area of 2000 forest lost in the year (km^2), in the 4,627 African cells with at least 1 km^2 of forest in 2000, 2001 to 2024, relative to 2001 to 2005, the reference period of equation (2) from the first year of the forest data. Panel A uses cell and country-by-year effects; Panel B replaces the country-by-year effects with country-by-year effects for each tercile of 2000 forest cover. Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5 percent. Gold-suitable land was therefore cleared faster in the very years when the higher price drew no workers back (Section 5.2).

In Table 3, I ask what kind of clearing this is. Sims et al. (2025) map the dominant driver of tree-cover loss over 2001 to 2024 for each square kilometre, and I split each year's loss by the driver of the pixel it falls in. Loss in pixels attributed to mining and energy infrastructure roughly doubles at the peak in a fully gold-suitable cell, by 0.024 log points on a mean of 0.021; it does not fall after the peak and rises by a further 0.013 in the second rise (column 2). Most of the additional clearing lies in pixels attributed to other drivers, such as permanent agriculture, shifting cultivation and logging, whose loss rises by about 21 percent at the peak and by a further 6 percent in the second rise (column 3). The map assigns one driver to each pixel for the whole period and is least accurate for the rare mining class, so this loss mixes land cleared for farms and settlements around the sites with mining that the map assigns to another driver. The second-rise clearing is on new land. It rises by 9 percent in cells with no record of gold or artisanal mining and not at all in cells with such a record, and it is concentrated in cells that cleared little forest in 2001 to 2005 (columns 4 to 7).

The household surveys record no mining equipment, so I turn to trade data for an indirect check at the level of countries. In Figure 6, I relate each African country's imports of mining machines and mercury, 1995 to 2024, to the gold-suitable share of its land in the same windows, with country and year effects. Relative to a country with no gold-suitable land, a country at Ghana's share of 0.42 imported about 2.4 times the value of excavators at the peak that it imported in 2000 to 2005, and these imports

Table 3—Forest loss by driver and by place

	All forested cells			Recorded gold or mining		Clearing in 2001–05	
	All loss	Mining pixels	Other pixels	Yes	No	High	Low
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Rise, 2006–11	0.072***	0.005*	0.071***	0.078***	0.060***	0.048*	0.085***
	(0.015)	(0.002)	(0.015)	(0.024)	(0.021)	(0.025)	(0.019)
Peak, 2012–14	0.197***	0.024***	0.194***	0.237***	0.162***	0.173***	0.178***
	(0.025)	(0.006)	(0.025)	(0.041)	(0.035)	(0.040)	(0.033)
After, 2015–18	0.188***	0.023***	0.188***	0.252***	0.144***	0.159***	0.173***
	(0.031)	(0.007)	(0.031)	(0.051)	(0.043)	(0.047)	(0.041)
Second rise, 2019–24	0.249***	0.036***	0.247***	0.240***	0.232***	0.188***	0.255***
	(0.032)	(0.008)	(0.032)	(0.051)	(0.045)	(0.046)	(0.044)
After minus peak	−0.009	−0.001	−0.007	0.016	−0.018	−0.014	−0.005
	(0.018)	(0.003)	(0.018)	(0.028)	(0.025)	(0.026)	(0.025)
Second rise minus after	0.061***	0.013***	0.059***	−0.012	0.088***	0.029	0.083***
	(0.014)	(0.004)	(0.014)	(0.023)	(0.019)	(0.021)	(0.019)
Mean of outcome	1.142	0.021	1.138	1.298	1.059	1.389	0.900
Cell-years	111,024	111,024	111,024	38,904	72,000	55,128	55,848
Cells	4,626	4,626	4,626	1,621	3,000	2,297	2,327

Notes: Each column reports equation (2), suitable × window, for the logarithm of one plus the area of 2000 forest lost in the year (km^2), in cells with at least 1 km^2 of forest in 2000, 2001 to 2024, relative to 2001 to 2005, with cell and country-by-year effects. Column 1 repeats Figure 5, Panel A. Columns 2 and 3 split each year's loss by the dominant driver of tree-cover loss in 2001 to 2024 mapped by Sims et al. (2025) at 1 km: column 2 counts loss in pixels whose driver is hard commodities (mining and energy infrastructure), column 3 loss in all other pixels. Columns 4 and 5 split cells by any record of gold or artisanal gold mining in the data of Girard et al. (2025) (Figure 2, Panel C); columns 6 and 7 by whether the cell's mean share of forest lost each year in 2001 to 2005 is above its country's median. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

stayed high after the peak and through the second rise. Mercury, the input of manual amalgamation, follows the price instead: its imports rise at the peak and fall back afterwards, as men's manual work does. Screening and washing machines and loaders do not respond, and road rollers, a construction machine I use as a placebo, rise after the peak, so the excavators that stayed may also serve a wider construction boom.

5.4 Human Development over the Cycle

Men move between farming and manual work as the price moves, and women move out of farming and out of work (Section 5.1). For schooling, living standards and health, the cycle leaves a longer trace. In Figure 7, I follow living standards, household wealth, night lights and child mortality through the same windows; Appendix Table B3 reports these estimates together with school attendance and three household amenities. Living standards and wealth come from households in the eight countries surveyed on both sides of the peak, and night lights and mortality from all countries with data.

For teenagers, the comparison can be made within a place. Children aged 6 to 11 in the same cell and survey would also feel a school closure, a mining camp or mercury,

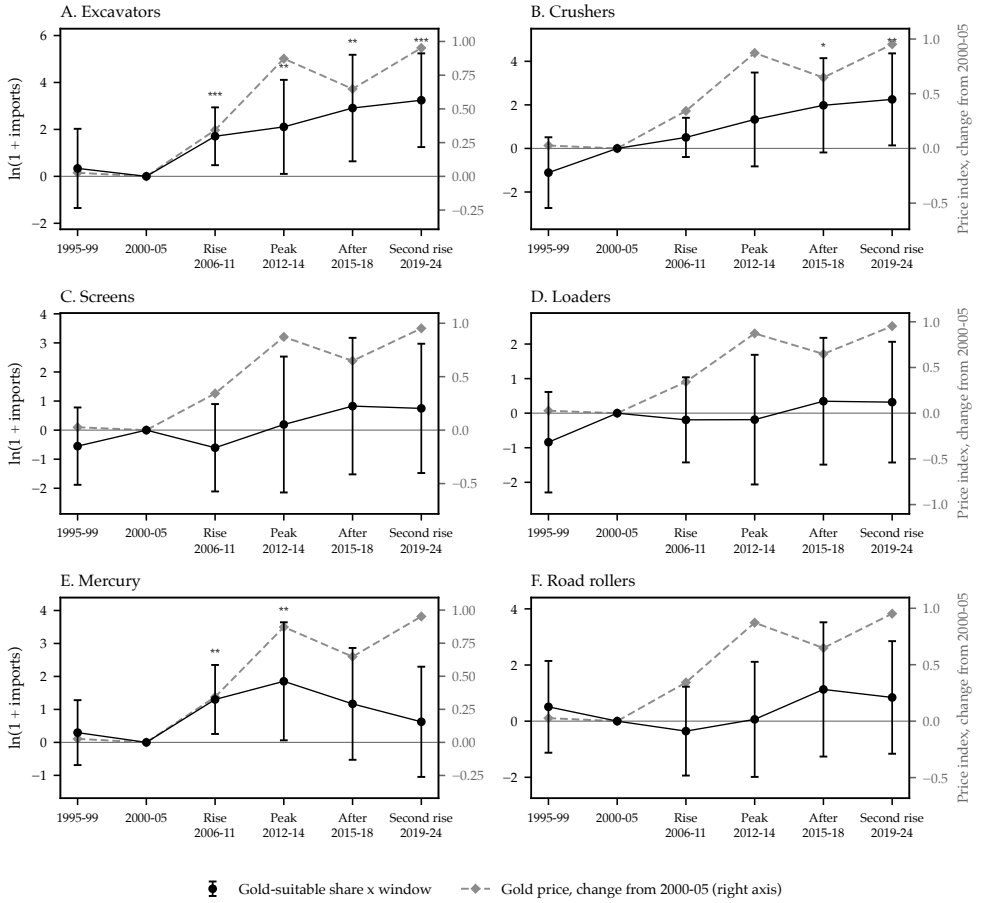


Figure 6—Imports of mining machines and mercury by country gold exposure

Notes: Coefficients of the window specification of equation (2) across African countries, relative to 2000 to 2005: the logarithm of one plus the value of the country's imports of each product in each year, 1995 to 2024, from CEPII BACI (Gaulier and Zignago, 2010), on the country's gold-suitable share times indicators for each window, with country and year effects and the logarithm of total imports, for 54 countries; Online Appendix Table OA4 reports the estimates, bootstrap p-values and the HS codes. The dashed line (right axis) is the mean lagged gold price index in each window minus its mean in 2000 to 2005 (2013 equal to one): 0.03 in 1995 to 1999, 0.34 during the rise, 0.87 at the peak, 0.65 after it and 0.95 in the second rise. Bars are 95 percent confidence intervals from standard errors clustered by country. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

but they are too young for the pits. I therefore estimate

$$D_{gcs} = \sum_{w \in \mathcal{W}} \beta_{wg} \left(S_c \times \mathbf{1}[t \in w] \right) + \alpha_{cg} + \lambda_{ktg} + u_{gcs}, \tag{3}$$

where D_{gcs} is the attendance rate of age group g (12 to 15, 16 to 20 or 21 to 24) minus that of children aged 6 to 11 in cell c and survey s, α_{cg} are cell-by-age-group effects and λ_{ktg} country-by-year-by-age-group effects. I weight by the number of children and start in 2000, because only one earlier survey records attendance.

In Figure 8, I plot the result for each age group. At the peak, attendance at ages 16 to 20 in a fully gold-suitable cell is 5.5 points lower than among younger children in

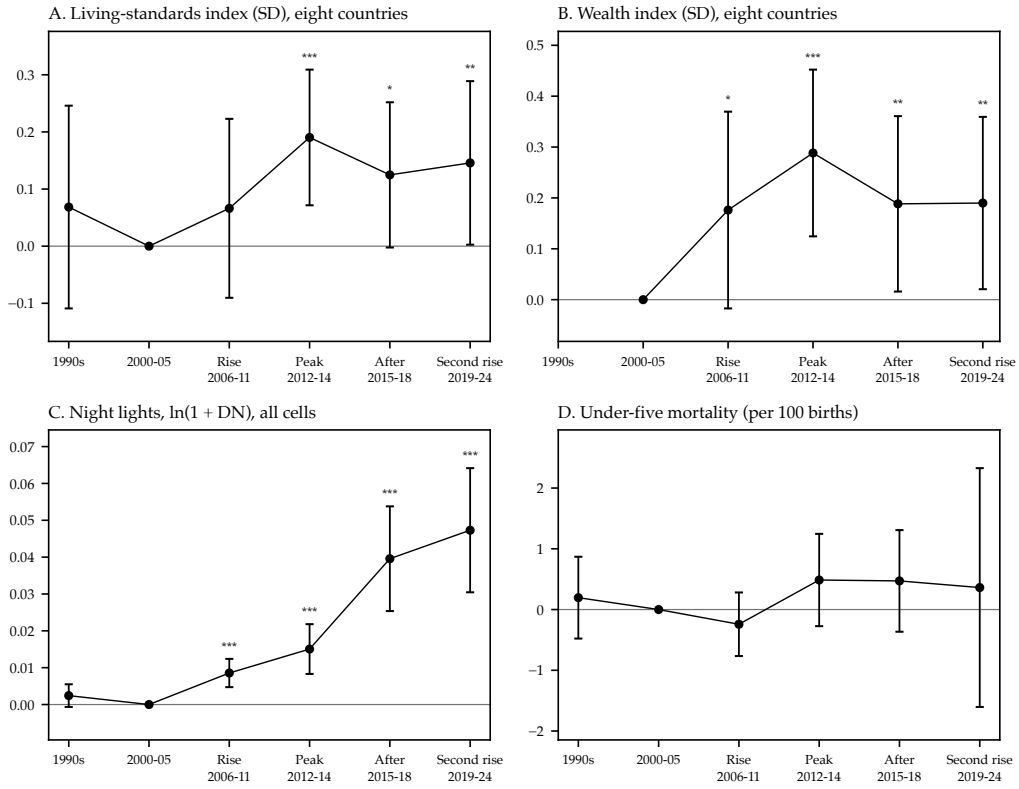


Figure 7—Living standards and health over the gold price cycle

Notes: Coefficients of equation (2) in a fully gold-suitable cell, relative to 2000 to 2005. Panel A is the living-standards index, the mean of the wealth index, electricity, improved water and (negatively) no toilet facility, each standardised on the households of 2000 to 2005; Panel B is the wealth index, standardised within each survey; both use households in the eight countries surveyed both at the peak and after it. Panel C is the logarithm of one plus mean harmonised night lights in all African cells, 1992 to 2024. Panel D is under-five mortality per 100 births, by year of birth. Appendix Table B3 reports the estimates, the specifications and Holm-adjusted p-values. Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the same place, against an attendance rate of 47 percent at those ages, and at ages 12 to 15 it is 3.1 points lower. Young adults aged 21 to 24, who have mostly left school, show no change, so the fall is specific to school-age teenagers. Boys and girls lose attendance alike (Figure 9, Panel A). Comparing attendance alone, the fall is larger where the household head has no schooling, 5.9 points against an attendance rate of 48 percent, compared with 2.6 points against 70 percent where the head has some; measured against younger children in the same place, it is similar for the two, at 6.1 and 5.3 points. Young men aged 15 to 19 leave farming at the peak, by 18 points on a mean of 32, and return to it after the peak, as older men do. Attendance returns only in part. The gap at ages 16 to 20 narrows by about half after the peak, to 3.0 points, and remains at 3.2 points in the second rise (Appendix Table B3, column 1). The lost attendance does not show up in completed schooling. Among household members aged 21 to 45, years of schooling do not fall with the price their cohort faced at ages 12 to 20: the estimate is -0.14 years on a mean of 5.9, with a 95 percent interval from

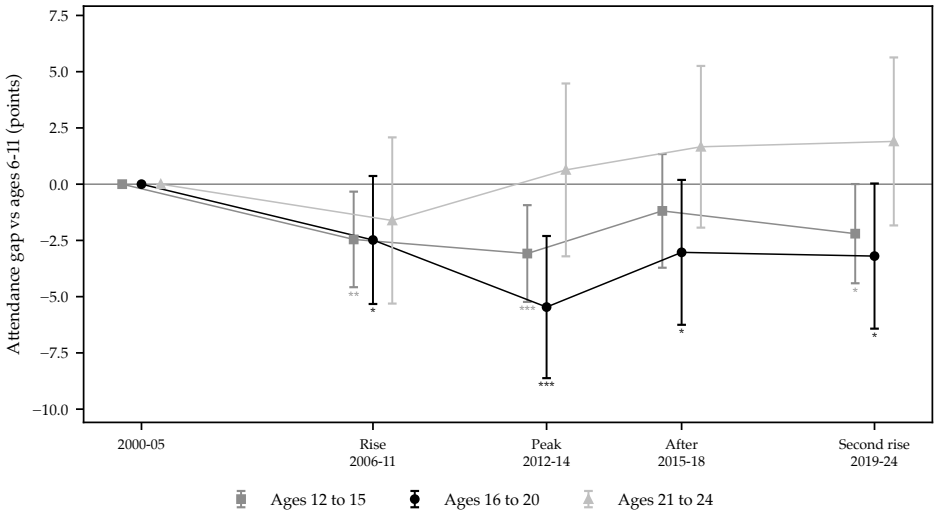


Figure 8—School attendance by age over the gold price cycle

Notes: Coefficients of equation (3): the attendance rate of each age group minus that of children aged 6 to 11 in the same cell and survey, in a fully gold-suitable cell, relative to 2000 to 2005, in percentage points (attendance is an indicator equal to 100 or 0). All countries with household rosters recording attendance, 2000 to 2025. Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

−0.36 to 0.09. Those born in 1990 to 1999, teenagers during the boom, have as much schooling relative to other cells as those born in 1985 to 1989.[7] The persistence I find is therefore a feature of places: teenagers in gold-suitable cells attend school less after the peak than before the boom, but the cohorts who were teenagers at the peak do not end up with less schooling.

Households keep most of what the boom brought. The wealth index, built from assets and housing, rises by 0.29 standard deviations at the peak in a fully gold-suitable cell (Figure 7, Panel B). When the price falls, it gives up 0.10 standard deviations, a third of its gain, and it holds the rest through 2019 to 2024; the second rise adds nothing to it, as it added nothing to men's work. Night lights tell the same story at the scale of the cell (Panel C). They brighten during the boom, and instead of dimming when the price falls they keep rising, so that gold-suitable cells are brighter after the peak and in the second rise than at the peak itself.

The household amenities move less cleanly. Electricity and improved drinking water are both higher at the peak than in 2000 to 2005 (Appendix Table B3, columns 6 and 7), but each already differed in the 1990s, electricity in the same direction and water in the opposite one, so the windows cannot attribute either to the boom. Households without a toilet facility do not change at the peak and fall in the second rise (column 8). Taken together, the index of the four rises by 0.19 standard deviations at the peak

[7] The cell is the place of residence at the interview, not in the teenage years. Schooling in gold-suitable cells has fallen behind that of other cells since the cohorts born in the 1960s, by 0.08 years per decade of birth, and no cohort of the boom departs from that trend; with the trend included, the estimate for the teenage price is 0.07 years.

and stays 0.12 above its level of 2000 to 2005 after it (Figure 7, Panel A); its fall after the peak, 0.07, is imprecise. Under-five mortality does not respond in any window (Panel D): the estimates are small relative to a mean of 11 deaths per 100 births. These results do not depend on how many outcomes I test. Adjusting for the four main outcomes, attendance, night lights, mortality and the living-standards index (Holm, 1979), the peak estimates for attendance, night lights and the living-standards index keep p-values below 0.01, and that for mortality is 0.21. Below, I show that child deaths rise in towns and not elsewhere.

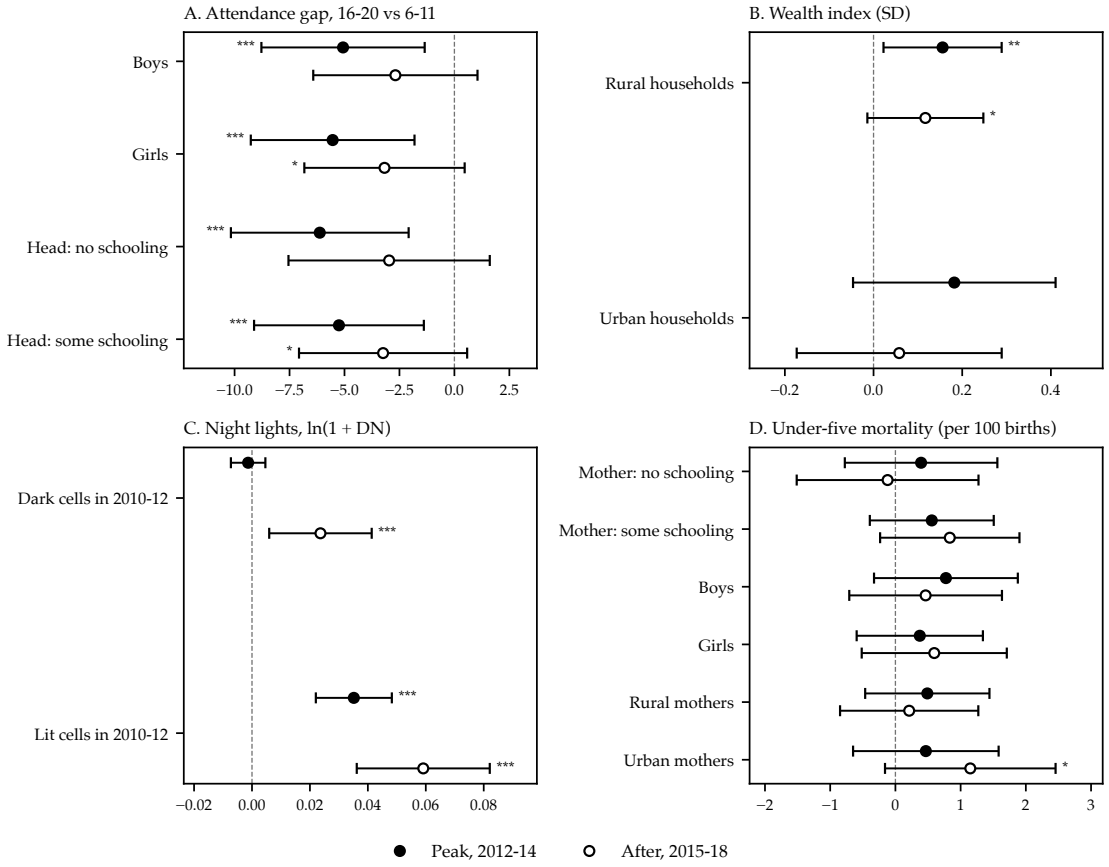


Figure 9—Human development over the gold price cycle, by group

Notes: Coefficients of equation (2) at the peak (2012 to 2014, filled) and after it (2015 to 2018, hollow), relative to 2000 to 2005, estimated separately for each group with the specifications of Table B3: the attendance gap of equation (3) by sex and by the schooling of the household head (Panel A), the wealth index for rural and urban households in the eight countries surveyed both at the peak and after it (Panel B), night lights for cells dark and lit in 2010 to 2012 (Panel C), and under-five mortality by the mother's schooling, the child's sex and the mother's urban or rural residence at interview (Panel D). Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Figure 9, I split these outcomes by group. Wealth rises among rural and urban households alike, by 0.16 and 0.18 standard deviations at the peak (Panel B). Both are smaller than the 0.29 for all households consistent with a boom that also moved households into towns: the share of surveyed households living in urban areas rises by 9 points at the peak, on a mean of 36 percent, consistent with the small towns that

artisanal gold created (Girard and Pignède, 2026). Light comes first to places that were already lit (Panel C). Cells lit before the boom brighten at the peak and again after it, while cells that were dark in 2010 to 2012 light up only after the peak. At the peak, under-five mortality does not respond for any group, whether split by the mother's schooling, the child's sex or residence (Panel D). After the peak, it is 1.1 points higher among children of urban mothers, on a mean of 8.4 deaths per 100 births, and 0.2 points higher among children of rural mothers. Residence at interview moves with the boom, so in Appendix Table B6, I instead split cells by whether they held an urban cluster in a survey fielded before 2006, close to the towns of 1975 that Girard and Pignède (2026) follow. In these towns, under-five mortality is 1.0 points higher at the peak and 1.2 points higher after it, on a mean of 10.5, and in the other cells it does not change. Like Girard and Pignède (2026), I find that child mortality rises with artisanal gold in towns, which the boom also made larger, and not in other cells.[8]

6 Robustness

In Figure 10, I re-estimate the cycle for the four outcomes that carry Section 5.1 under ten changes to the specification, and I report the estimates in Appendix Table B4. The first row repeats Appendix Table B1. Measuring exposure as any gold-suitable land in the cell, as a suitable share of at least one half, or as the share of the cell within 10 km of suitable land rescales the coefficients but leaves the cycle in place: men's manual work and women's sales rise at the peak and fall after it, and men's farming falls and recovers. Controlling for industrial gold mines times the price, whether those producing by 2000 or every mine, moves no estimate by more than 0.8 points, and neither does dropping the temperature and crop controls. Pooling 1990 to 2005 as the reference period lowers the peak coefficients, to 4.5 points for men's manual work, -13.9 for their farming and 4.0 for women's sales, because the 1990s already differed, but it leaves every change after the peak within 0.1 points. Without the clusters whose displacement could have moved them across a cell edge, the fall in men's manual work after the peak is 3.1 points and the recovery of farming 6.2.

The cycle in work is not produced by people moving in at the peak and out after it. Among respondents who have always lived in the place or have lived there for at least five years, men's manual work rises by 8.2 points at the peak and falls by 5.8 after it, and men's farming falls by 16.0 points and recovers 13.0. In Appendix Table B5, I estimate equation (2) for who lives in the surveyed households. At the peak, households in gold-suitable cells are larger by 0.4 people on a mean of 5.9, the share of young men is 1.0 point higher and the share of women 0.8 points lower, and none of these changes after

[8]The rise is smaller and imprecise for deaths before age one: 0.6 points at the peak and 0.8 after it in the towns, on a mean of 6.5. Defining towns as cells lit at night in 1992 to 1999 gives 0.7 points at the peak, imprecisely, and 1.0 after it, but that definition counts 83 percent of births as urban.

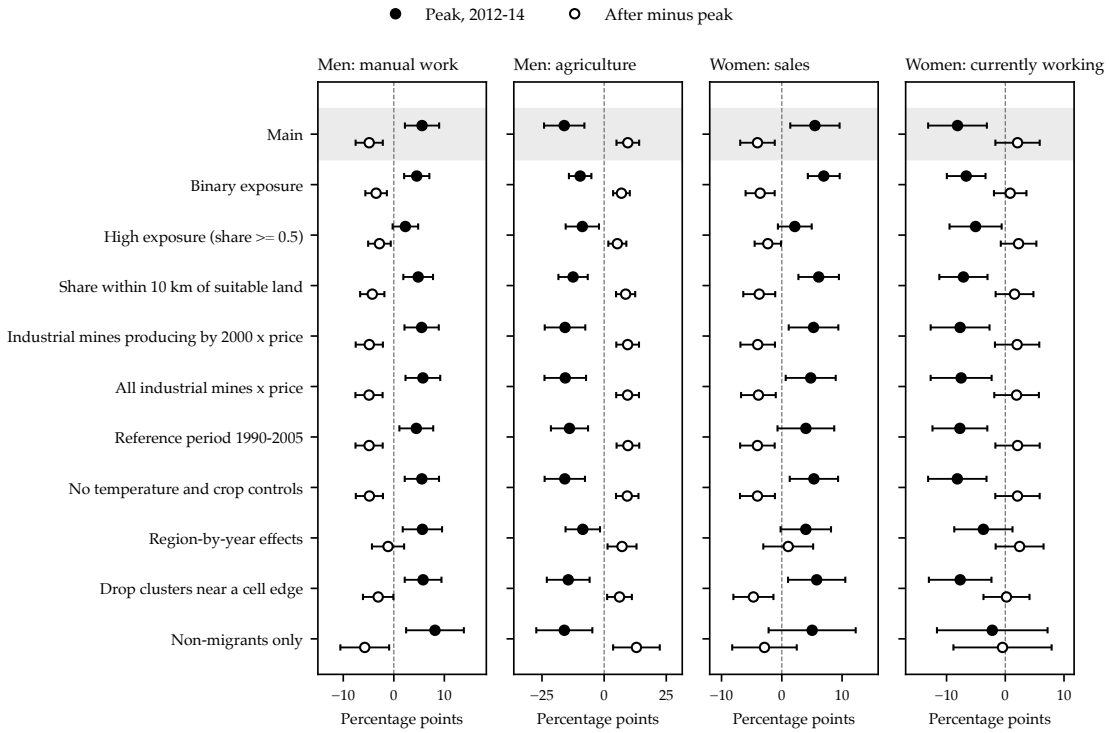


Figure 10—Robustness of the cycle estimates

Notes: Each row reports equation (2) for the outcome in the panel title with the change named in the row, in the eight countries surveyed both at the peak and after it, surveys to 2018, in percentage points for a fully gold-suitable cell. Filled circles are the coefficient on suitable × peak (2012–14), relative to 2000 to 2005; hollow circles are the change from the peak to 2015–18. The shaded row is the specification of Appendix Table B1. The specification with a suitable × linear trend is reported only in Appendix Table B4, because its intervals are about three times as wide. Bars are 95 percent confidence intervals from standard errors clustered by cell. Appendix Table B4 reports the estimates and defines each row.

the peak. Households were already larger in gold-suitable cells before the boom, by 1.4 people in the 1990s, and the cell fixed effects absorb that level. Women who arrived in the previous five years are no more common at the peak; they are 6.5 points more common after it, when manual work had already fallen. The one outcome that differs among stayers is women's current work, whose fall at the peak is 2.2 points against 8.1 in the full sample, so part of women's withdrawal from work may come from women who moved.

Two checks change the result. With region-by-year effects, which compare each cell only with others in the same survey region, men's manual work still rises by 5.7 points at the peak and men's farming still falls and recovers, by 8.6 and 7.2 points, but manual work falls by only 1.1 points after the peak and women's sales no longer fall. Part of the reversal in manual work and trade is therefore common to whole gold regions, while the reversal in farming holds within them. Adding the suitable share times a linear trend leaves the windows without precision: in eight countries the trend is separated from the windows only by the timing of surveys within each window, the standard errors of the peak coefficients triple, and the implied trend for men's manual work is

−12.7 points a decade.

Dropping any one of the eight countries leaves the rise of men's manual work at the peak between 4.8 and 7.3 points and its fall after the peak between 3.2 and 5.7 points; farming recovers by 7.9 to 10.6 points in every sample. With eight countries, inference across countries is weak.[9] The placebo does not depend on the number of countries. In 1,000 reassignments of the suitable share across cells within each country, no more than 2 percent of draws give a rise at the peak or a reversal after it as large as the actual estimates, for men's manual work and farming and for women's sales, and no more than 0.1 percent give a fall in women's current work as large as the actual one (Appendix Table B4).

Online Appendix OA2 re-estimates Table 2 and the cycle with exposure measured as the share of the cell within 10 km of gold-suitable land (see also Ghazarian and Khan, 2025), and the estimates change little: the cycle in men's manual work and women's sales is within a point of the main estimates, and the fall in men's farming at the peak is smaller, 12.5 points against 16.1.

7 Conclusion

Between 2000 and 2012, the gold price rose sixfold; by 2015 it had fallen by 30 percent, and from 2019 it rose above its earlier peak. I follow work, land and human development in Africa's gold-suitable places through this cycle, using 166 household surveys linked to the geology that artisanal mining needs. The jobs lasted as long as the price. In the eight countries surveyed on both sides of the peak, men in a fully gold-suitable cell took up manual work and left farming as the price rose, and returned to the fields when it fell; women moved into trade and back out of it, and fewer women worked at all. When the price rose again after 2019, the jobs did not return, while forest loss grew further.

Other changes outlast the price. Forest clearing in gold-suitable cells did not slow when the price fell, so the land taken by the boom kept growing. Households kept about two thirds of their gain in wealth, and the night lights of gold-suitable cells kept rising after the peak. Teenagers aged 16 to 20 lost 5.5 points of school attendance relative to younger children in the same place, and half of that loss remained after the jobs had gone, yet the cohorts who were teenagers during the boom completed as much schooling as those before them. Child mortality rose in the towns that predate the boom.

The asymmetry matters for how artisanal mining is judged. Employment at the peak is a poor measure of what a gold rush leaves behind: the work is gone a few years

[9] Wild cluster restricted bootstrap p-values with country clusters, over all 256 Rademacher draws, are 0.02 for the fall of men's farming at the peak and 0.08 for the rise of women's sales, and between 0.12 and 0.51 for the other six statistics in Appendix Table B4.

later, and the cleared forest, the lower school attendance of gold-suitable places and the assets households bought remain. Formalisation programmes and crackdowns act on the part of the activity that already follows the price, while the cleared forest stays cleared and half of the lost school attendance does not return when the price falls. The second rise cleared more forest, much of it on land with no earlier record of mining, and drew no workers back. Excavator imports in gold-exposed countries stayed high after the peak while mercury imports followed the price, which fits a turn to machines, but road rollers rose too and the new clearing lies away from mapped mines. The household surveys record neither equipment nor the scale of operations, and the trade data cannot place a machine at a site. Repeated satellite maps of mining pits, or licence registers, would tell whether the next gold rush employs anyone at all.

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A Data

A.1 Household Surveys

I assemble the Demographic and Health Surveys from the raw recode files of every standard survey in 33 African countries: 166 surveys fielded between 1986 and 2025 (Table A1). From each survey I keep the women's, men's, household, household-member and birth recodes, and I link each sampling cluster to its half-degree cell through the survey's GPS file. The 132 surveys with coordinates locate 65,746 clusters, and 99 percent of the people interviewed in them are linked to a cell. Ethiopian surveys record dates in the Ethiopian calendar, and I convert them to the Gregorian calendar by adding 92 months to the century-month codes, except in the 2024 survey, which already uses Gregorian dates.

Online Appendix OA1 compares this build with the survey file of Girard et al. (2025): of their 189 survey-years, 145 are in this build, and in those I match 97.6 percent of their women one to one.

B Additional Results

This appendix reports the estimates behind Figures 3, 4, 7 and 10 and the checks cited in Sections 5 and 6.

Table A1—Demographic and Health Surveys by country

Country	Survey years	Women	Men	Households	Fall	Rise
Angola	2015, 2023	29,096	12,288	32,352		Yes
Benin	1996, 2001, (2006), 2012, 2017	43,731	16,792	41,376	Yes	
Burkina Faso	1993, 1999, 2003, 2010, 2021	58,909	22,670	45,803		
Burundi	(1987), 2010, 2016	26,591	11,815	24,517		
Cameroon	1991, (1998), 2004, 2011, 2018	44,540	20,200	39,817		Yes
Chad	(1997), (2004), 2014	17,719	5,248	17,233	Yes	
Congo Democratic Republic	2007, 2013, 2023	54,755	25,305	51,706		
Cote d'Ivoire	1994, 1998, 2012, 2021	35,816	16,028	32,217		
Egypt	(1988), 1992, 1995, 2000, 2003, 2005, 2008, 2014	104,735	0	119,674		
Eswatini	2006	4,908	4,086	4,756		
Ethiopia	2000, 2005, 2011, 2016, 2019, 2024	90,463	46,911	89,614		Yes
Gabon	(2000), 2012, 2019	19,408	12,477	21,419		
Ghana	(1988), 1993, 1998, 2003, 2008, 2014, 2022	44,200	23,693	59,269		
Guinea	1999, 2005, 2012, 2018	34,608	13,012	26,302	Yes	
Kenya	(1989), (1993), (1998), 2003, 2008, 2014, 2022, (2025)	79,680	34,223	91,710		
Lesotho	2004, 2009, 2014, 2023	27,297	12,072	36,599		
Liberia	1986, 2007, 2013, 2019	29,390	14,213	24,967		
Madagascar	(1992), 1997, (2004), 2008, 2021	42,993	17,476	45,244		
Malawi	(1992), 2000, 2004, 2010, 2015, 2024	93,538	29,755	101,528		Yes
Mali	(1987), 1996, 2001, 2006, 2012, 2018, 2023	77,746	26,666	62,580	Yes	Yes
Mozambique	(1997), (2003), 2011, 2022	26,910	9,410	28,149		
Namibia	(1992), 2000, 2006, 2013	26,409	11,285	25,277		
Niger	1992, 1998, (2006), 2012	25,240	9,040	21,920		
Nigeria	1990, 2003, 2008, 2013, 2018, 2024	169,021	60,489	168,706	Yes	Yes
Rwanda	(1992), (2000), 2005, 2008, 2010, 2015, 2019, 2025	74,499	37,120	70,048	Yes	Yes
Senegal	(1986), 1993, 1997, 2005, 2010, 2012, 2014, 2015, 2016, 2017, 2018, 2019, 2023	130,745	45,261	66,514	Yes	Yes
Sierra Leone	2008, 2013, 2019	38,990	17,487	32,816		
South Africa	(1998), 2016	8,514	3,618	11,083		
Tanzania	(1992), (1996), 1999, (2004), 2010, 2015, 2022	42,234	15,206	41,112		Yes
Togo	1988, 1998, 2013	21,378	8,283	17,027		
Uganda	(1988), (1995), 2000, 2006, 2011, 2016	40,960	11,543	43,293		
Zambia	(1992), (1996), (2002), 2007, 2013, 2018, 2024	50,884	45,732	48,441	Yes	Yes
Zimbabwe	(1988), (1994), 1999, 2005, 2010, 2015	33,389	25,258	35,366		
Total	166 surveys, 132 with GPS	1,649,296	664,662	1,578,435	8	10

Notes: Standard DHS surveys in Africa, 1986 to 2025, listed by the year of the survey's name. Years in parentheses are surveys without cluster coordinates, which cannot be linked to cells and do not enter the estimates. Women, men and households are interviews linked to a cell. Fall marks the eight countries with interviews both at the peak (2012–14) and after it (2015–18), by interview date, which can fall in the year after the survey's name; rise marks the ten countries surveyed both in 2015–18 and in 2019–24.

Table B1—Work over the gold price cycle

	Working (1)	Manual (2)	Extraction (3)	Agriculture (4)	Sales (5)	Services (6)
Panel A: All adults						
Suitable × 1990s	−2.25	−0.91		−6.03	2.82	0.47
	(2.29)	(1.19)		(3.68)	(2.50)	(0.59)
Suitable × rise, 2006–11	−8.41***	−0.26	0.88	−6.86**	2.62*	−1.80***
	(2.46)	(1.25)	(0.62)	(2.78)	(1.56)	(0.64)
Suitable × peak, 2012–14	−7.50***	2.15*	1.09	−12.86***	4.51***	−1.36*
	(2.46)	(1.19)	(0.70)	(2.79)	(1.57)	(0.77)
After minus peak	1.25	−2.42***	−0.83*	5.24***	−2.78***	−0.13
	(1.60)	(0.85)	(0.45)	(1.97)	(1.04)	(0.82)
Mean of outcome (%)	63.6	10.1	0.9	29.6	19.7	4.2
Observations	637,014	620,509	408,237	620,509	620,509	620,509
Panel B: Men						
Suitable × 1990s	−3.73	3.80		−7.08	−1.31	−1.62
	(3.89)	(2.40)		(4.42)	(1.49)	(1.20)
Suitable × rise, 2006–11	−9.30**	2.49	1.32**	−9.66**	1.41	−1.83*
	(3.93)	(1.85)	(0.56)	(4.55)	(1.50)	(0.95)
Suitable × peak, 2012–14	−4.56	5.58***	1.24**	−16.06***	2.18	−0.65
	(3.93)	(1.73)	(0.58)	(4.14)	(1.39)	(0.73)
After minus peak	−1.62	−4.87***	−0.50	9.52***	−0.48	−1.56**
	(1.91)	(1.37)	(0.54)	(2.33)	(0.86)	(0.74)
Mean of outcome (%)	79.0	19.0	1.3	40.8	10.0	4.6
Observations	181,812	185,152	133,282	185,152	185,152	185,152
Panel C: Women						
Suitable × 1990s	−1.15	−2.96**		−4.34	4.26	1.06*
	(2.60)	(1.35)		(3.96)	(3.10)	(0.56)
Suitable × rise, 2006–11	−7.52***	−1.17	0.76	−5.79*	3.18	−1.85***
	(2.63)	(1.51)	(0.77)	(2.96)	(2.07)	(0.66)
Suitable × peak, 2012–14	−8.14***	0.73	1.11	−11.02***	5.50***	−1.65*
	(2.55)	(1.37)	(0.89)	(3.03)	(2.09)	(0.96)
After minus peak	2.09	−0.88	−1.01*	2.84	−4.04***	0.59
	(1.93)	(1.04)	(0.60)	(2.17)	(1.46)	(1.06)
Mean of outcome (%)	57.4	6.3	0.7	24.8	23.8	4.0
Observations	455,201	435,356	274,954	435,356	435,356	435,356

Notes: Each column reports equation (2) for an indicator equal to 100 if the respondent's own occupation is in the sector and 0 otherwise, so that coefficients are in percentage points, in the eight countries surveyed both at the peak and after it (Benin, Chad, Guinea, Mali, Nigeria, Rwanda, Senegal and Zambia), surveys to 2018. Coefficients are the change in a fully gold-suitable cell relative to 2000 to 2005, in percentage points; the window 2015–18 enters the regression and is reported as its difference from the peak. Panel A stacks the men's and women's interviews and controls for sex. Extraction is manual work whose occupation label names mining or quarrying, in the surveys whose labels code it. All regressions control for temperature, the crop-price index, age and age squared, with cell and country-by-year effects. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B2—Work in the second rise of the gold price

		2019–24 minus 2015–18 (1)	Implied by the peak response (2)	Mean (3)	Observations (4)
All adults	Manual work	0.1	3.6	12.0	1,072,390
		(1.0)			
	Agriculture	1.6	−7.5	29.1	1,072,390
		(1.8)			
	Sales	−0.6	3.1	15.9	1,072,390
		(1.0)			
Men	Manual work	−0.2	9.3	21.4	323,711
		(1.7)			
	Agriculture	2.2	−14.6	39.5	323,711
		(2.3)			
Women	Currently working	−1.1	−2.8	53.8	768,233
		(1.9)			
	Sales	−0.4	5.3	18.8	748,678
		(1.3)			

Notes: Column 1 is the difference between the 2019–24 and 2015–18 coefficients of equation (2) in a fully gold-suitable cell, in percentage points, for each respondent's own occupation, as an indicator equal to 100 or 0, in the 10 countries surveyed both in 2015–18 and in 2019–24, surveys to 2024, with age controls (and sex for all adults) and without the temperature and crop controls, which end in 2018. Column 2 scales the change from 2015–18 to the peak, estimated in the same regression and countries, by the change in the mean lagged price index at the interview dates, 1.13 at the peak, 0.88 in 2015–18 and 1.28 in 2019–24. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B3—Human development over the gold price cycle

	All countries			Eight countries surveyed at and after the peak				
	Attendance 16–20 vs 6–11 (1)	Night lights ln(1 + DN) (2)	Under-five mortality (3)	Living-standards index (SD) (4)	Wealth index (SD) (5)	Electricity (6)	Improved water (7)	No toilet facility (8)
Suitable × price, eq. (1)	−2.5*	0.042***	0.33	0.11**	0.12**	4.4*	3.9	−4.6**
	(1.3)	(0.008)	(0.46)	(0.05)	(0.06)	(2.3)	(2.8)	(2.2)
Suitable × window, eq. (2)								
1990s		0.002	0.20	0.07		9.9***	−9.7**	3.1
		(0.002)	(0.34)	(0.09)		(3.0)	(4.3)	(5.3)
Rise, 2006–11	−2.5*	0.009***	−0.24	0.07	0.18*	2.6	7.6*	3.0
	(1.5)	(0.002)	(0.27)	(0.08)	(0.10)	(3.3)	(4.1)	(2.7)
Peak, 2012–14	−5.5***	0.015***	0.49	0.19***	0.29***	7.3***	9.8***	−1.3
	(1.6)	(0.003)	(0.39)	(0.06)	(0.08)	(2.8)	(3.6)	(2.2)
After, 2015–18	−3.0*	0.040***	0.47	0.12*	0.19**	8.8***	4.9	−0.2
	(1.6)	(0.007)	(0.43)	(0.06)	(0.09)	(2.9)	(3.6)	(2.3)
Second rise, 2019–24	−3.2*	0.047***	0.36	0.15**	0.19**	6.9**	5.1	−4.5
	(1.6)	(0.009)	(1.00)	(0.07)	(0.09)	(3.2)	(3.7)	(2.9)
After minus peak	2.4*	0.025***	−0.01	−0.07	−0.10**	1.5	−4.9**	1.1
	(1.4)	(0.005)	(0.33)	(0.05)	(0.04)	(2.3)	(2.3)	(1.7)
Second rise minus after	−0.2	0.008*	−0.11	0.02	0.00	−2.0	0.1	−4.3*
	(1.4)	(0.004)	(0.97)	(0.06)	(0.05)	(2.6)	(2.8)	(2.2)
Holm-adjusted p, columns 1 to 4								
Suitable × price	[0.12]	[0.00]	[0.47]	[0.10]				
Peak	[0.00]	[0.00]	[0.21]	[0.00]				
After minus peak	[0.24]	[0.00]	[0.96]	[0.32]				
Second rise minus after	[1.00]	[0.25]	[1.00]	[1.00]				
Mean of outcome	−23.0	0.192	11.30	0.26	0.00	38.1	65.2	25.5
Observations	33,853	352,044	1,948,162	491,641	447,602	491,661	490,049	490,522
Countries	33	54	33	8	8	8	8	8

Notes: The first row reports equation (1) and the rows below it equation (2); column 1 uses equation (3). Column 1 is the attendance rate of household members aged 16 to 20 minus that of members aged 6 to 11 in the same cell and survey, in percentage points, in all countries with household rosters recording attendance, from 2000; observations are cell-survey-age-group means, weighted by the number of children. Column 2 is the logarithm of one plus mean harmonised night lights in all African cells, 1992 to 2024, with country-by-year effects for each 2010–12 brightness group. Column 3 is an indicator equal to 100 if a child died before age five, for births in the 15 years before interview and at least 60 months before it, by year of birth, with cell-by-survey and country-by-birth-year effects and controls for the mother's age at birth and its square, sex and twin birth. Columns 4 to 8 use households in the eight countries surveyed both at the peak and after it: the wealth index, standardised within each survey, and indicators equal to 100 or 0 for electricity, an improved source of drinking water (piped, borehole, protected well or spring, rainwater or bottled) and no toilet facility (columns 5 to 8). The living-standards index (column 4) is the mean of the four, each standardised by its mean and standard deviation among households surveyed in 2000 to 2005 and no toilet facility entering with a negative sign, for households with at least three of them, divided by its standard deviation in 2000 to 2005 (Kling et al., 2007; Anderson, 2008). Holm p-values adjust for testing four outcomes, columns 1 to 4, separately for each row (Holm, 1979). Surveys to 2024, without the temperature and crop controls. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B4—Robustness of the cycle estimates

	Men		Women	
	manual work (1)	agriculture (2)	sales (3)	working (4)
Panel A: Suitable × peak, 2012–14				
Main	5.58*** (1.73)	−16.06*** (4.14)	5.50*** (2.09)	−8.14*** (2.55)
Binary exposure	4.54*** (1.27)	−9.65*** (2.31)	6.96*** (1.35)	−6.66*** (1.68)
High exposure (share ≥ 0.5)	2.29* (1.29)	−8.78** (3.42)	2.16 (1.43)	−5.06** (2.26)
Within 10 km of suitable land	4.82*** (1.50)	−12.50*** (3.01)	6.12*** (1.71)	−7.13*** (2.09)
Industrial mines by 2000 × price	5.51*** (1.73)	−15.74*** (4.16)	5.26** (2.10)	−7.70*** (2.56)
All industrial mines × price	5.75*** (1.75)	−15.64*** (4.27)	4.79** (2.12)	−7.52*** (2.65)
Suitable × linear trend	18.28*** (5.34)	−19.89 (12.49)	5.78 (4.34)	9.93 (6.68)
Reference period 1990–2005	4.45*** (1.71)	−13.94*** (3.81)	3.99* (2.40)	−7.73*** (2.38)
No temperature, crop controls	5.55*** (1.73)	−15.83*** (4.14)	5.32*** (2.05)	−8.17*** (2.54)
Region-by-year effects	5.65*** (1.99)	−8.60** (3.54)	3.98* (2.13)	−3.73 (2.53)
No clusters near a cell edge	5.79*** (1.85)	−14.45*** (4.39)	5.78** (2.42)	−7.68*** (2.72)
Non-migrants only	8.15*** (2.92)	−16.02*** (5.75)	5.03 (3.69)	−2.22 (4.81)
Drop one country, range	[4.8, 7.3]	[−19.1, −12.3]	[3.3, 6.8]	[−11.8, −5.2]
Wild bootstrap p, countries	0.12	0.02	0.08	0.16
Placebo permutation p	0.005	<0.001	0.009	0.001
Panel B: After minus peak				
Main	−4.87*** (1.37)	9.52*** (2.33)	−4.04*** (1.46)	2.09 (1.93)
Binary exposure	−3.50*** (1.09)	6.97*** (1.71)	−3.60*** (1.23)	0.83 (1.41)
High exposure (share ≥ 0.5)	−2.84** (1.14)	5.31*** (1.85)	−2.32** (1.12)	2.26 (1.55)
Within 10 km of suitable land	−4.26*** (1.22)	8.64*** (1.96)	−3.75*** (1.35)	1.57 (1.64)
Industrial mines by 2000 × price	−4.85*** (1.37)	9.46*** (2.33)	−4.00*** (1.46)	2.02 (1.93)
All industrial mines × price	−4.91*** (1.37)	9.43*** (2.34)	−3.88*** (1.47)	1.92 (1.94)
Suitable × linear trend	1.12 (2.87)	7.72 (5.41)	−3.90 (2.57)	10.72*** (2.95)
Reference period 1990–2005	−4.89*** (1.37)	9.56*** (2.33)	−4.05*** (1.46)	2.09 (1.93)
No temperature, crop controls	−4.84*** (1.37)	9.29*** (2.30)	−4.04*** (1.48)	2.08 (1.93)
Region-by-year effects	−1.14 (1.62)	7.19** (2.98)	1.06 (2.11)	2.44 (2.08)
No clusters near a cell edge	−3.10** (1.54)	6.19** (2.54)	−4.71*** (1.69)	0.21 (2.00)
Non-migrants only	−5.75** (2.46)	13.00*** (4.79)	−2.88 (2.73)	−0.47 (4.27)
Drop one country, range	[−5.7, −3.2]	[7.9, 10.6]	[−5.4, −2.2]	[0.3, 5.8]
Wild bootstrap p, countries	0.18	0.16	0.30	0.51
Placebo permutation p	0.005	<0.001	0.019	0.384

Notes: Each row reports equation (2) for the respondent's own occupation or current work, in percentage points, in the eight countries surveyed both at the peak and after it, surveys to 2018, with the change named in the row. The first row is the specification of Appendix Table B1: temperature, the crop-price index, age and age squared, with cell and country-by-year effects. Binary exposure is any gold-suitable land in the cell; high exposure is a suitable share of at least 0.5. Industrial mines are cells with an industrial gold mine producing by 2000, or at any date, times the lagged price. The trend is the suitable share times decades since 2000. With the 1990–2005 reference, the 1990s window is dropped. Region-by-year effects use the survey regions. Clusters near a cell edge lie within the DHS displacement distance of it. Non-migrants have always lived in the place or for at least five years. Dropping one country reports the range of estimates over the eight samples. Bootstrap p-values are from the wild cluster restricted bootstrap with Rademacher weights over all 256 draws. Placebo p-values are the share of 1,000 reassignments of the suitable share across cells within country whose estimate exceeds the actual one in absolute value, estimated on cell-by-year means. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5—Who is surveyed over the gold price cycle

	1990s (1)	Rise 2006–11 (2)	Peak 2012–14 (3)	After minus peak (4)	Mean (5)
Household size	1.39***	0.63**	0.40	−0.04	5.85
	(0.35)	(0.28)	(0.27)	(0.10)	
Men 15–29 (% of members)	−0.2	0.9	1.0	0.4	12.6
	(0.9)	(0.7)	(0.7)	(0.5)	
Female (% of members)	0.5	−2.0**	−0.8	−0.1	50.1
	(0.8)	(0.8)	(0.7)	(0.5)	
Head with no schooling (%)	−2.1	−1.3	−3.3*	1.8	47.0
	(2.7)	(2.2)	(1.9)	(1.3)	
Women arrived <5 years ago (%)	5.7***	−0.1	−1.5	6.5**	19.5
	(2.2)	(2.8)	(3.7)	(3.0)	

Notes: Each row reports equation (2) in the eight countries surveyed both at the peak and after it, surveys to 2018, with cell and country-by-year effects and no controls. The first four rows use households from the household rosters; the last uses women aged 15 to 49, with an indicator equal to 100 if she has lived in the place for less than five years. Coefficients are the change in a fully gold-suitable cell relative to 2000 to 2005. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B6—Child mortality in towns that predate the boom

	Lit at night in 1992–99				Urban cluster before 2006			
	Towns		Other cells		Towns		Other cells	
	Under five (1)	Infant (2)	Under five (3)	Infant (4)	Under five (5)	Infant (6)	Under five (7)	Infant (8)
Rise, 2006–11	−0.05	−0.27	−0.75	−0.74	0.25	0.06	−0.52	−0.69**
	(0.30)	(0.26)	(0.62)	(0.56)	(0.38)	(0.32)	(0.37)	(0.33)
Peak, 2012–14	0.70	0.32	−0.45	−0.39	1.03**	0.58	−0.03	−0.18
	(0.43)	(0.37)	(0.91)	(0.77)	(0.52)	(0.44)	(0.56)	(0.47)
After, 2015–18	1.01**	0.64	−1.07	−0.25	1.20**	0.78	0.01	0.15
	(0.46)	(0.41)	(1.04)	(0.85)	(0.61)	(0.55)	(0.60)	(0.50)
After minus peak	0.31	0.31	−0.62	0.15	0.18	0.20	0.04	0.34
	(0.36)	(0.31)	(0.79)	(0.59)	(0.48)	(0.44)	(0.45)	(0.35)
Mean	11.0	6.7	12.9	7.7	10.5	6.5	12.1	7.3
Births	1,608,991	1,608,991	339,165	339,165	1,013,257	1,013,257	934,899	934,899
Cells	2,573	2,573	1,899	1,899	858	858	3,614	3,614

Notes: Each column reports equation (2), suitable × window, for an indicator equal to 100 if the child died before age five (under five) or age one (infant), by year of birth, for births in the 15 years before interview and at least 60 months before it, in all countries, with cell-by-survey and country-by-birth-year effects and controls for the mother's age at birth and its square, the child's sex and twin birth. Columns 1 to 4 define towns as cells with positive mean harmonised night lights in 1992 to 1999; columns 5 to 8 as cells with at least one urban DHS cluster in a survey fielded before 2006. The mean is deaths per 100 births. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix

OA1 The Survey File of Girard et al. (2025)

The survey file of Girard et al. (2025) covers 189 survey-years. Of these, 145 are in this build; the other 44, which hold 15 percent of the women in their file, are mostly malaria and AIDS indicator surveys, which this build excludes. In the shared survey-years, I match 97.6 percent of their women one to one on survey, year and month of interview, age and cell, and across surveys the number of women per cell correlates at a median of 0.994.

In Table OA1, I re-estimate their specification for the measure that counts the woman's or her partner's occupation. On the new build and their survey-years, manual work rises by 3.8 points per unit of the price index, against 2.8 on their file, extraction by 1.1 on both, and agriculture falls by 2.5 against 2.8 (rows 1 and 2). Services differ, at 0.8 against 3.0 points. The new build holds more women in the same survey-years, 1,154,443 against 869,054, and keeping one woman per household brings the count to 824,370 and the estimate for services to 1.4 points (row 3). Adding the surveys to 2018 that their file does not contain leaves the estimates close to row 2 (row 4).

Table OA1—Reproducing Girard et al. (2025) on the new survey build

	Manual (1)	Extraction (2)	Agriculture (3)	Services (4)	Women
Girard et al. file	2.84**	1.06	−2.77	3.02**	869,054
	(1.18)	(0.73)	(1.71)	(1.39)	
Same survey-years	3.82***	1.09**	−2.49	0.76	1,154,443
	(1.12)	(0.44)	(1.84)	(1.50)	
One woman per household	4.12***	1.16**	−2.55	1.42	824,370
	(1.24)	(0.47)	(1.85)	(1.56)	
All surveys to 2018	3.88***	1.08**	−2.48	1.18	1,177,619
	(1.09)	(0.43)	(1.83)	(1.49)	

Notes: Each entry is the coefficient on the suitable share times the lagged gold price index (2013 equal to one) for an indicator equal to 100 if the woman or her partner works in the sector, the outcome of Girard et al. (2025), in their specification: cell and country-by-year effects, temperature, crop suitability times crop prices, industrial gold times price, age and age squared, surveys to 2018. Sectors follow the DHS occupation groups of the woman's (v717) and her partner's (v705) occupation; extraction is manual work whose occupation label names mining or quarrying, in the surveys whose labels code it. The first row uses their survey file; rows 2 to 4 use the survey files assembled for this paper. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA2 Exposure within 10 km of Gold-Suitable Land

In Table OA2, I re-estimate Table 2 and Appendix Table B1 with exposure measured as the share of the cell within 10 km of gold-suitable land (see also Ghazarian and Khan,

2025). Across all countries, agriculture falls by 5.2 points per unit of the price index, against 4.9 in Table 2, and women's sales rise by 4.1 points, against 3.8. In the eight countries surveyed on both sides of the peak, men's manual work rises by 4.8 points at the peak and gives up 4.3 after it, men's farming falls by 12.5 points and recovers 8.6, and women's sales rise by 6.1 points and fall by 3.8.

Table OA2—Work with exposure within 10 km of gold-suitable land

	Working (1)	Manual (2)	Extraction (3)	Agriculture (4)	Sales (5)	Services (6)
Panel A: All adults						
Exposure × price, all countries	−2.27*	0.89	−0.04	−5.24***	3.06***	−0.94**
	(1.16)	(0.69)	(0.30)	(1.42)	(0.86)	(0.40)
Exposure × peak, fall countries	−6.16***	2.24**	0.77	−11.77***	4.93***	−1.45**
	(1.89)	(1.00)	(0.49)	(2.16)	(1.31)	(0.66)
After minus peak, fall countries	0.71	−2.10***	−0.81**	4.77***	−2.87***	0.00
	(1.38)	(0.77)	(0.38)	(1.67)	(0.97)	(0.70)
Mean of outcome, all countries	61.0	10.3	0.9	31.1	14.0	4.2
Panel B: Men						
Exposure × price, all countries	−1.23	1.13	−0.12	−5.11***	0.54	−0.39
	(1.42)	(0.95)	(0.53)	(1.53)	(0.65)	(0.53)
Exposure × peak, fall countries	−2.49	4.82***	0.99**	−12.50***	1.89	−0.44
	(2.76)	(1.50)	(0.41)	(3.01)	(1.21)	(0.67)
After minus peak, fall countries	−2.10	−4.26***	−0.60	8.64***	−1.08	−1.64**
	(1.70)	(1.22)	(0.47)	(1.96)	(0.75)	(0.68)
Mean of outcome, all countries	77.3	18.4	1.9	41.2	7.6	5.2
Panel C: Women						
Exposure × price, all countries	−2.65*	1.10	0.18	−5.22***	4.13***	−1.17**
	(1.42)	(0.77)	(0.31)	(1.64)	(1.14)	(0.48)
Exposure × peak, fall countries	−7.13***	1.17	0.75	−10.86***	6.12***	−1.86**
	(2.09)	(1.12)	(0.63)	(2.39)	(1.71)	(0.81)
After minus peak, fall countries	1.57	−0.79	−0.94*	2.57	−3.75***	0.82
	(1.64)	(0.92)	(0.51)	(1.87)	(1.35)	(0.87)
Mean of outcome, all countries	54.7	7.0	0.5	27.0	16.5	3.8

Notes: The first row of each panel reports equation (1) and the next two rows equation (2), with exposure equal to the share of the cell within 10 km of gold-suitable land in place of the suitable share. The first row uses all countries; the other two use the eight countries surveyed both at the peak and after it. Surveys to 2018, own occupations and current work from the men's and women's interviews, indicators equal to 100 or 0. All regressions control for temperature, the crop-price index, age and age squared, with cell and country-by-year effects; Panel A also controls for sex. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA3 Employment by Sector in the Censuses

In Table OA3, I estimate equation (1) on the IPUMS-International censuses (Minnesota Population Center, 2026), with the suitable share of each census unit in place of the cell's. No country has a census both at the peak and after it, so the censuses cannot follow the cycle. Thirteen countries have at least two rounds from 1985, in 1,097 districts or regions. Men's employment in mining rises by 0.8 points per unit of the price index, on a mean share of 1.8 percent, in line with the rise of extraction in the surveys. Agriculture does not fall: its share of workers rises by 2.8 points, and that of services

falls by 3.7, and with country clusters neither estimate is statistically significant. The censuses classify workers by industry, not by occupation, and over districts and regions much larger than a cell, and a digger who also farms may be counted in agriculture.

Table OA3—Artisanal gold and employment by sector in the censuses

	Agriculture (1)	Mining (2)	Manufacturing (3)	Services (4)
All workers: suitable × price	2.82*	0.46	−0.18	−3.69**
	(1.50)	(0.29)	(0.49)	(1.44)
p, country clusters	[0.12]	[0.33]	[0.75]	[0.17]
Mean share (%)	45.2	1.3	8.6	44.3
Men: suitable × price	1.32	0.84**	−0.14	−3.18**
	(1.53)	(0.37)	(0.45)	(1.36)
p, country clusters	[0.46]	[0.28]	[0.77]	[0.27]
Mean share (%)	41.5	1.8	10.3	45.8
Women: suitable × price	3.67**	0.02	−0.03	−3.87**
	(1.69)	(0.22)	(0.67)	(1.55)
p, country clusters	[0.03]	[0.88]	[0.98]	[0.07]
Mean share (%)	51.6	0.5	5.8	41.8
Unit-rounds, all workers	3,115	3,115	3,115	3,115
Units	1,097	1,097	1,097	1,097
Countries	13	13	13	13

Notes: Each entry reports equation (1) for the share of workers in the sector, in percent, in census units (districts where the census identifies them, otherwise regions) of the IPUMS-International censuses from 1985, in countries with at least two rounds. Exposure is the suitable share of the unit times the lagged gold price index (2013 equal to one). Services include trade. Unit and country-by-year effects, weighted by the number of workers. Standard errors clustered by unit in parentheses; p-values from country clusters with $t(G-1)$ critical values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, from unit clusters.

OA4 Imports of Mining Machines and Mercury

In Table OA4, I report the estimates behind Figure 6, the window specification of equation (2) across African countries, with bootstrap p-values. The outcome is the logarithm of one plus the value of the country's imports of each product in each year, 1995 to 2024, from CEPII BACI (Gaulier and Zignago, 2010), which reconciles the reports of importers and exporters. Exposure is the gold-suitable share of the country, the area-weighted mean of the cell shares, which ranges from 0 to 0.74 and is 0.42 in Ghana. The regressions include country and year effects and the logarithm of total imports; standard errors are clustered by country, for 54 countries. Excavators rise during the rise and at the peak and do not fall afterwards (column 1); a coefficient of 2.11 at the peak implies that a country at Ghana's share imported 2.4 times its imports of 2000 to 2005, relative to a country without gold-suitable land. Crushing machines follow the same path, imprecisely (column 2), while screening and washing machines and loaders do

Table OA4—Imports of mining machines and mercury by country gold exposure

	Excavators (1)	Crushers (2)	Screens (3)	Loaders (4)	Mercury (5)	Road rollers (6)
1995–99	0.34	−1.11	−0.55	−0.84	0.30	0.51
	(0.86)	(0.83)	(0.68)	(0.74)	(0.50)	(0.83)
Rise, 2006–11	1.71***	0.51	−0.61	−0.19	1.30**	−0.36
	(0.63)	(0.46)	(0.77)	(0.63)	(0.53)	(0.81)
Peak, 2012–14	2.11**	1.33	0.19	−0.19	1.85**	0.06
	(1.02)	(1.10)	(1.19)	(0.96)	(0.91)	(1.05)
p, bootstrap	[0.04]	[0.23]	[0.87]	[0.87]	[0.04]	[0.95]
After, 2015–18	2.91**	1.98*	0.83	0.35	1.17	1.13
	(1.16)	(1.10)	(1.20)	(0.94)	(0.87)	(1.22)
Second rise, 2019–24	3.24***	2.25**	0.75	0.32	0.62	0.84
	(1.02)	(1.08)	(1.13)	(0.89)	(0.85)	(1.02)
After minus peak	0.80	0.65	0.63	0.53	−0.68	1.07*
	(0.85)	(0.75)	(0.56)	(0.50)	(0.78)	(0.63)
p, bootstrap	[0.33]	[0.42]	[0.27]	[0.31]	[0.36]	[0.11]
Second rise minus after	0.33	0.27	−0.08	−0.03	−0.54	−0.29
	(0.71)	(0.63)	(0.61)	(0.45)	(0.56)	(0.75)
p, bootstrap	[0.66]	[0.67]	[0.90]	[0.95]	[0.33]	[0.70]
Mean of outcome	7.48	6.96	5.96	7.95	1.17	6.78
Country-years	1,578	1,578	1,578	1,578	1,578	1,578
Countries	54	54	54	54	54	54

Notes: Each column regresses the logarithm of one plus the value of imports (thousand current USD) by African country and year, 1995 to 2024, from CEPII BACI (Gaulier and Zignago, 2010), on the country's gold-suitable share (the area-weighted mean of the cell share of equation (1), from 0 to 0.74) times indicators for the windows of equation (2), relative to 2000 to 2005, with country and year effects and the logarithm of total imports. HS 1992 codes: excavators with a revolving superstructure 842952; crushing and grinding machines 847420; sorting, screening, separating and washing machines 847410; front-end loaders 842951; mercury 280540; tamping machines and road rollers 842940. Standard errors clustered by country in parentheses; wild cluster restricted bootstrap *p*-values (999 Rademacher draws) in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, from country-clustered standard errors.

not respond (columns 3 and 4). Mercury rises during the rise and at the peak and falls back afterwards (column 5); exporter bans from 2011 and the Minamata Convention, in force from 2017, push the recorded trade down after the peak as well. Road rollers do not respond at the peak but rise after it (column 6). The estimates change little without South Africa, with quantities in place of values, or with exposure above the median as a binary indicator; without the control for total imports, road rollers rise at the peak as well.